

Week 03. Variance. Bi-variate data. Correlation and Causation.

Part 1: Statistics and Econometrics. MK68-2024/25

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- Cause, Effect, Potential Outcome, and the Ceteris Paribus
- Ceteris Paribus: Public vs. Private University
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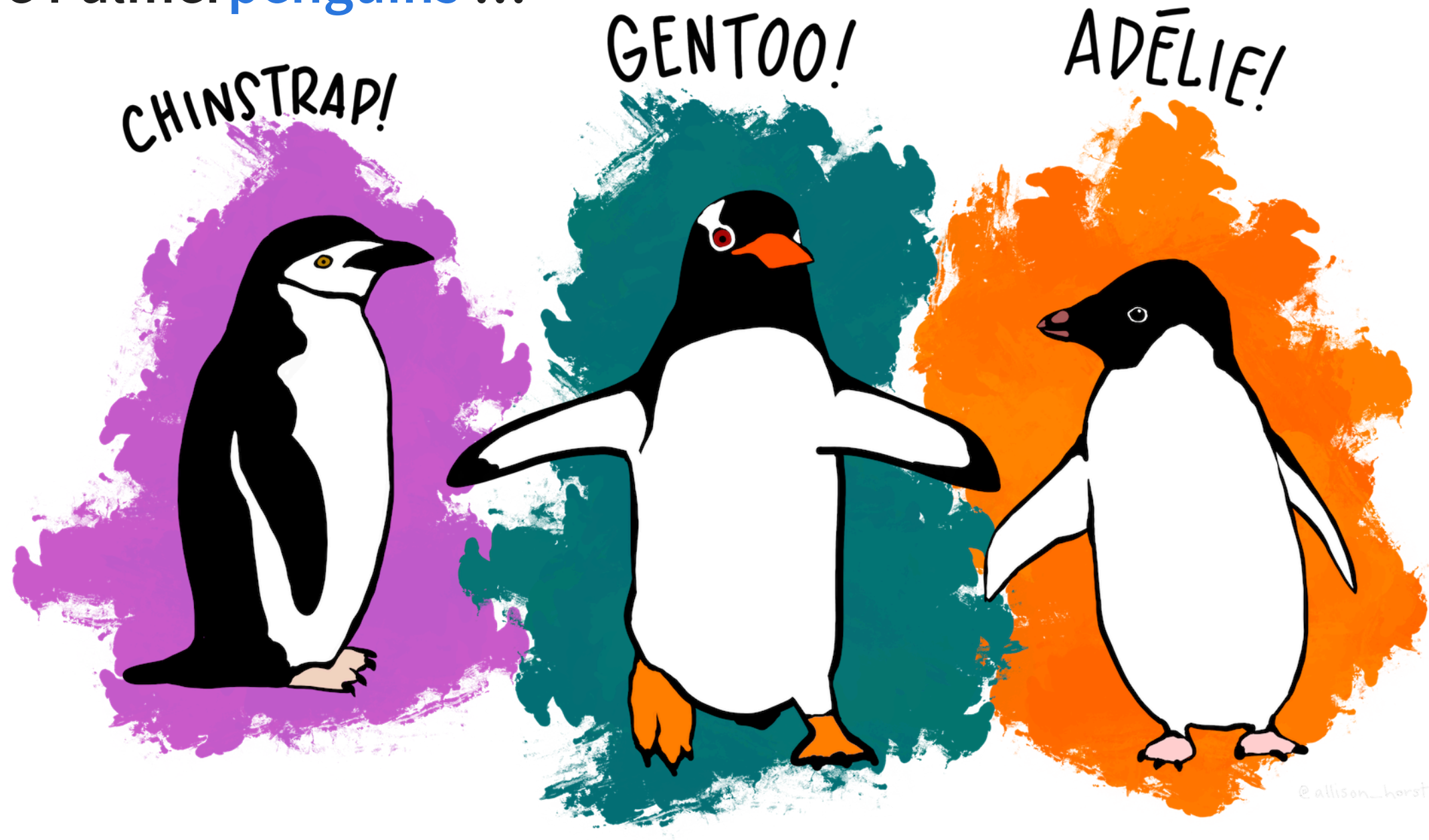
Bi-variate data

Data on two variables:

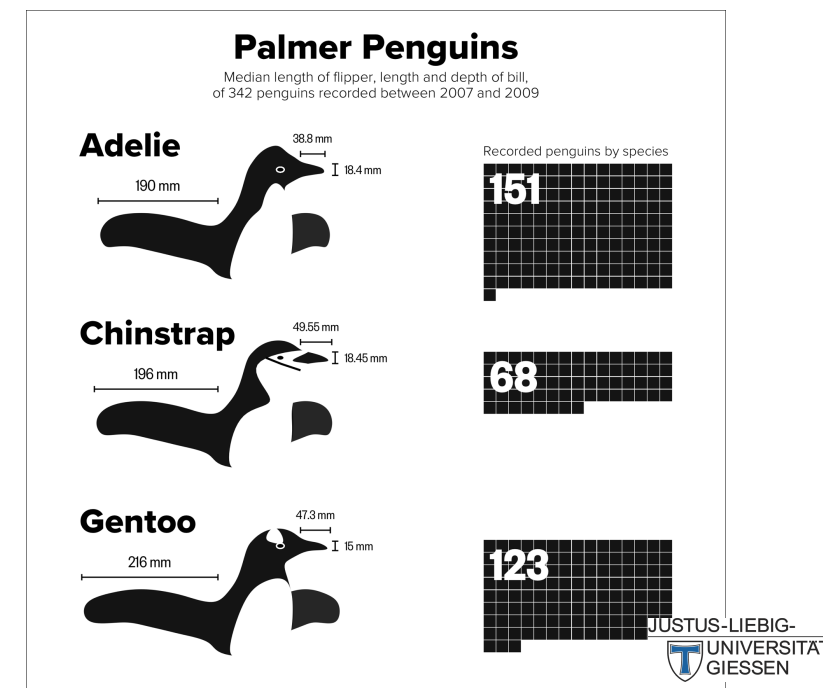
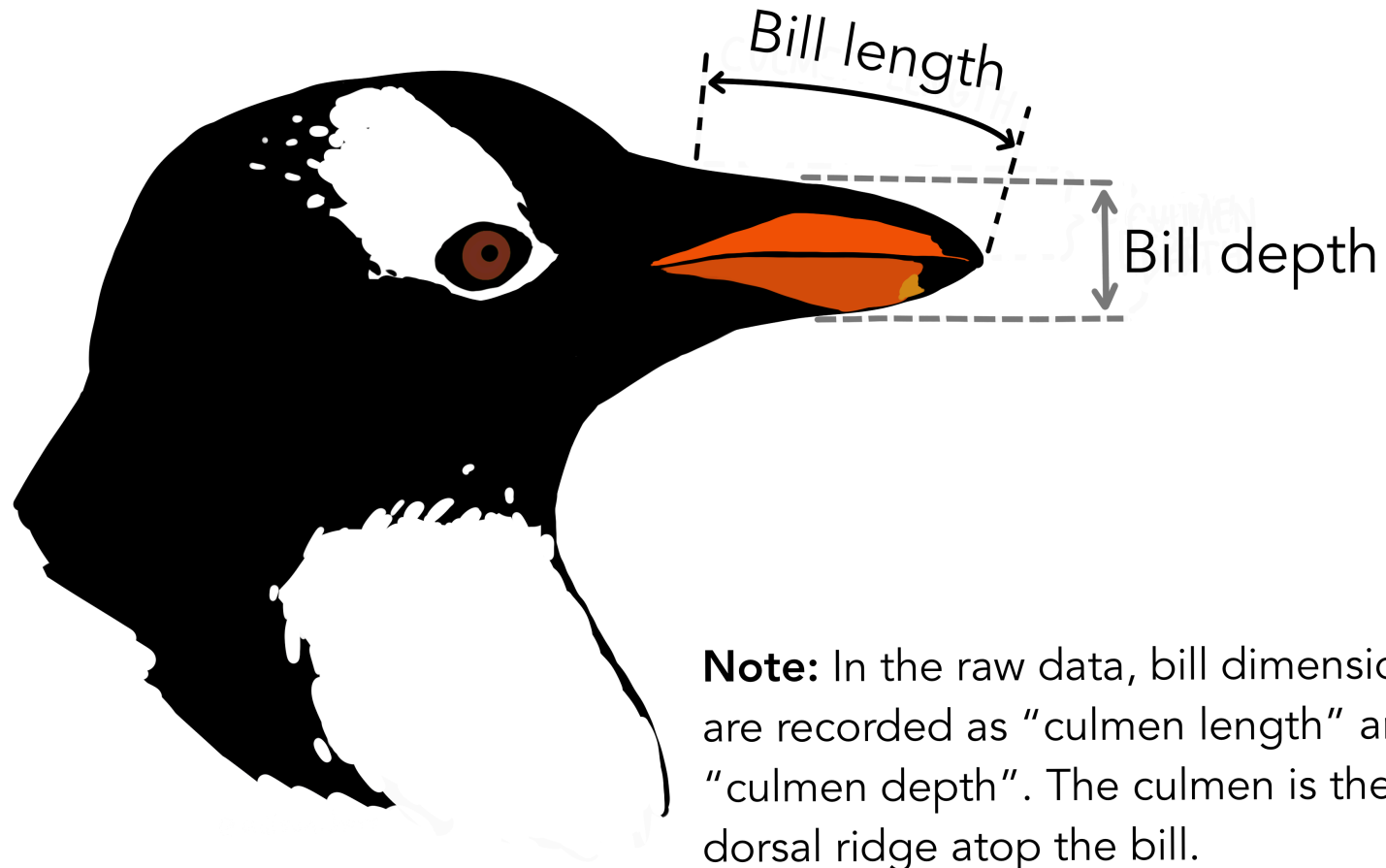
- Where each variable is paired with another.

Example of Bi-variate data... (1)

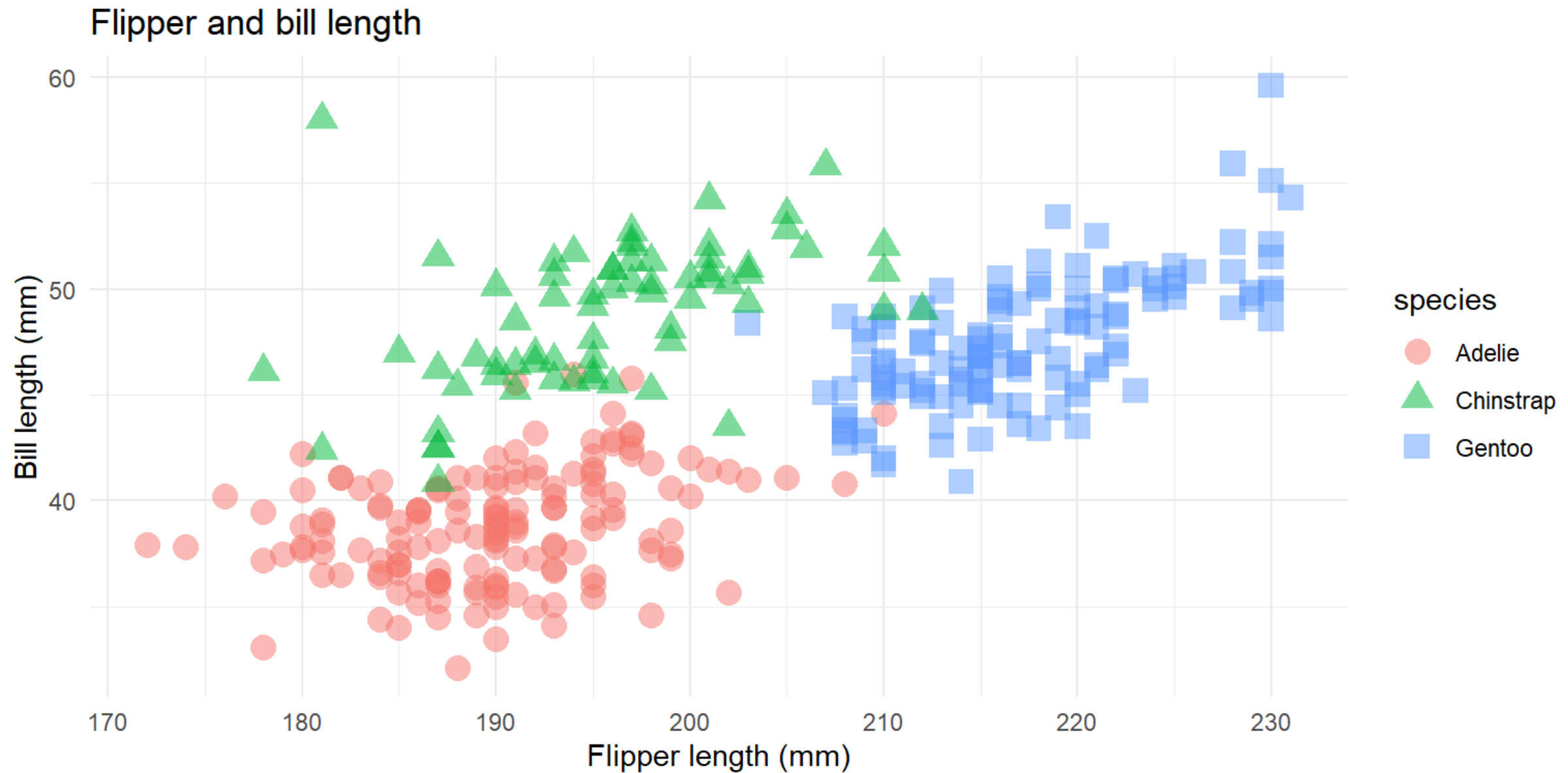
are Palmer **penguins** !!!



Example of Bi-viriate data... (2)



Building a scatter plot with penguins



Scatter plot

Type of a plot that shows the relationship between two numeric variables.

Convention:

- x is the horizontal axis and y is the vertical;
- x changes first and then y follows;
- if any causality is implied, x causes y ;

More information:

- Khan Academy. [Unit: Exploring bivariate numerical data. Introduction to scatter plots](#)

Good scatter plot is self explaining

It contains:

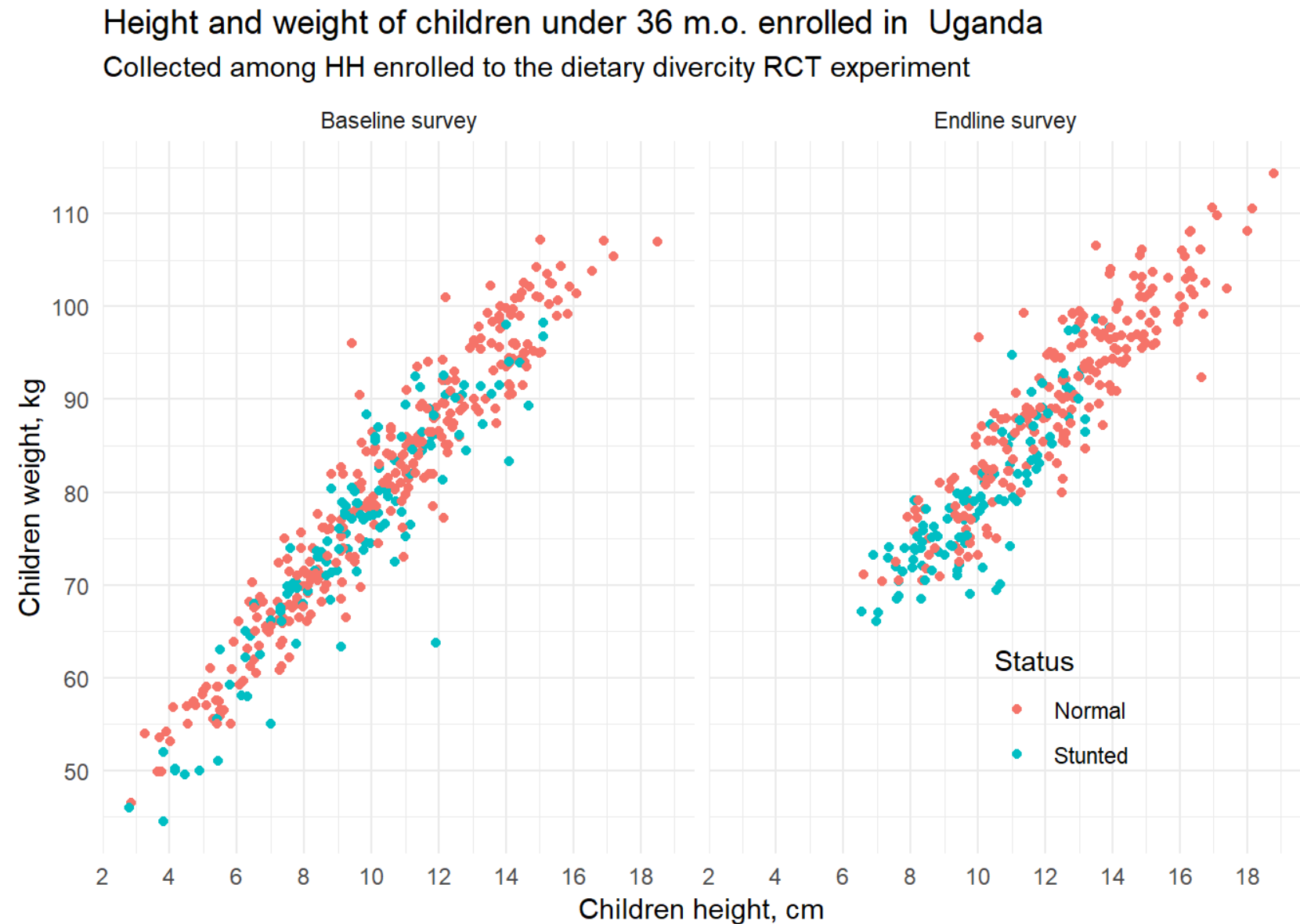
Title and sources

Axis labels, units

Points

Legends

Temporal dimension



Source: Jordan I, Röhlig A, Glas MG, Waswa LM, Mugisha J, Krawinkel MB, Nuppenau E-A. Dietary Diversity of Women across Agricultural Seasons in the Kapchorwa District, Uganda: Results from a Cohort Study. *Foods*. 2022; 11(3):344. <https://doi.org/10.3390/foods11030344>

Trend line

Is the line that best outlines a linear trend in the relationship between x and y .

More information:

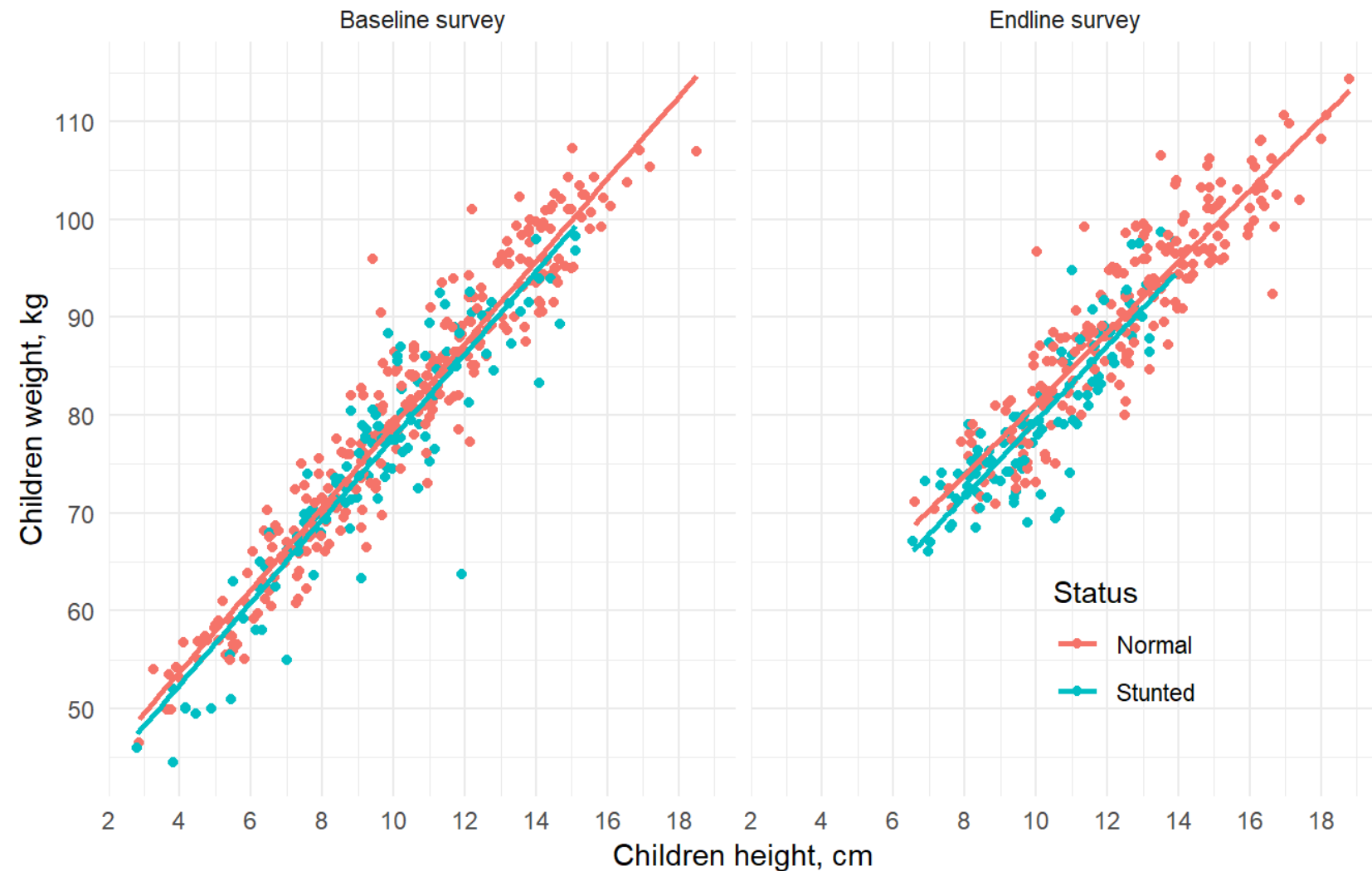
- Khan Academy. [Unit: Exploring bivariate numerical data. Introduction to trend lines](#)

Trend line

Trend line is also a

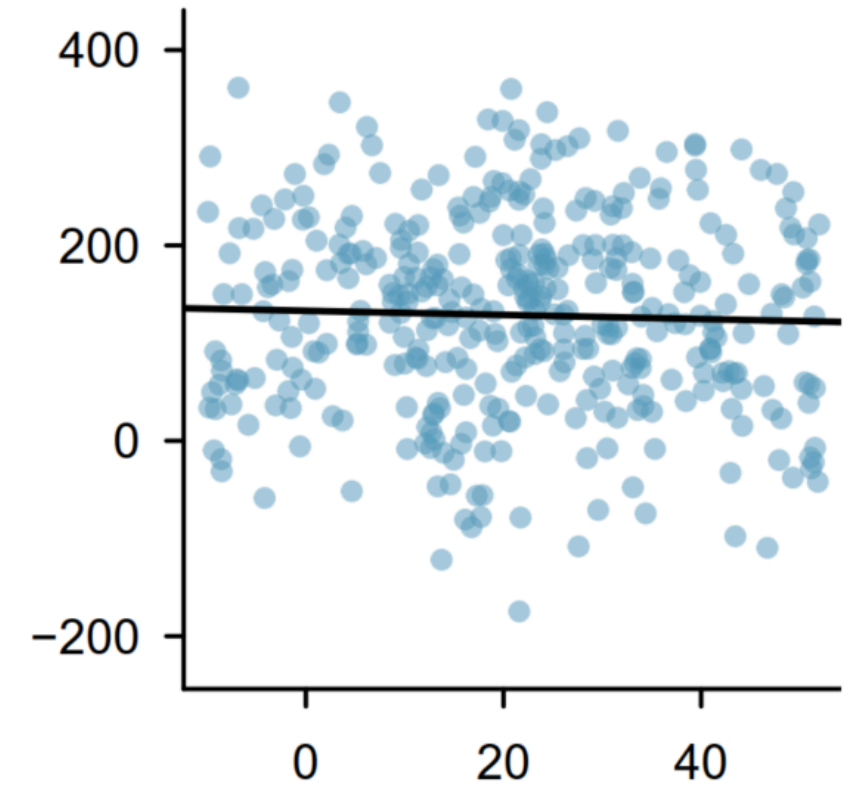
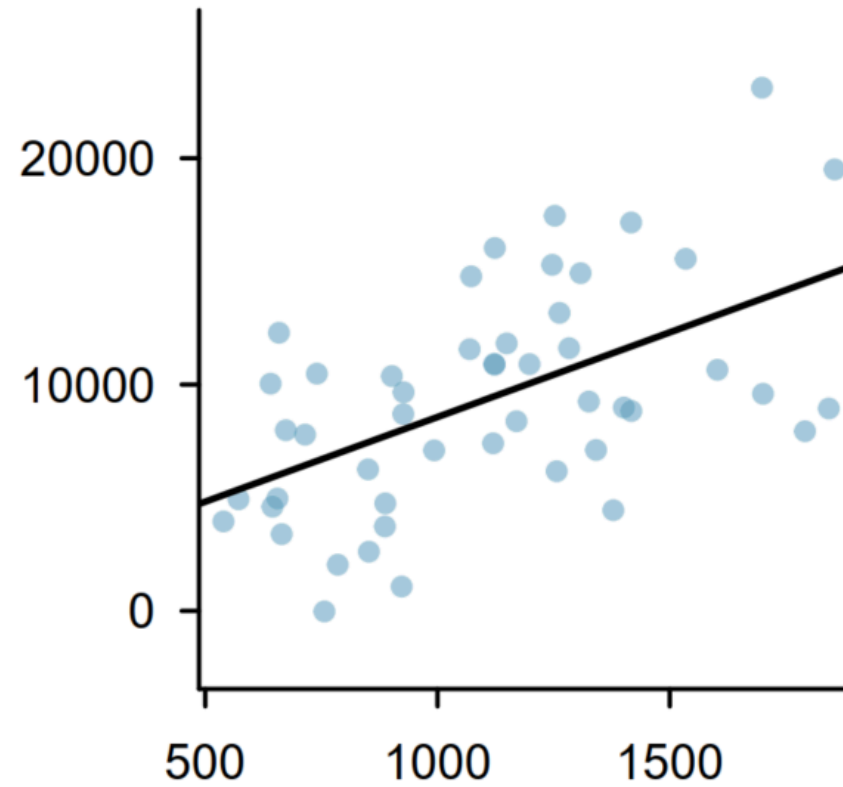
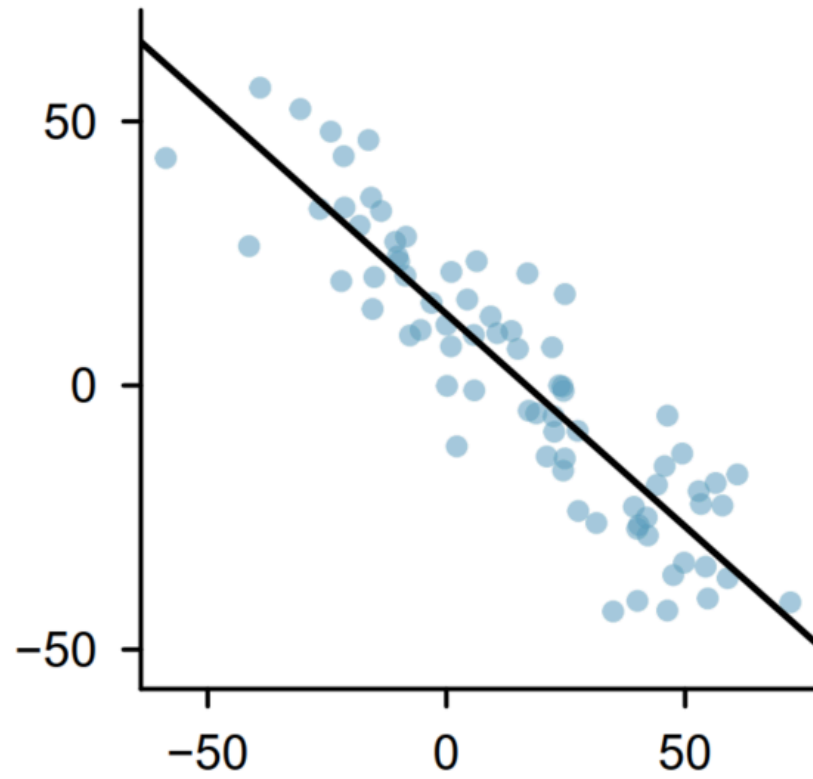
- Simple linear regression line

Height and weight of children under 36 m.o. enrolled in Uganda
Collected among HH enrolled to the dietary diversity RCT experiment



Source: Jordan I, Röhlrig A, Glas MG, Waswa LM, Mugisha J, Krawinkel MB, Nuppenau E-A. Dietary Diversity of Women across Agricultural Seasons in the Kapchorwa District, Uganda: Results from a Cohort Study. *Foods*. 2022; 11(3):344. <https://doi.org/10.3390/foods11030344>

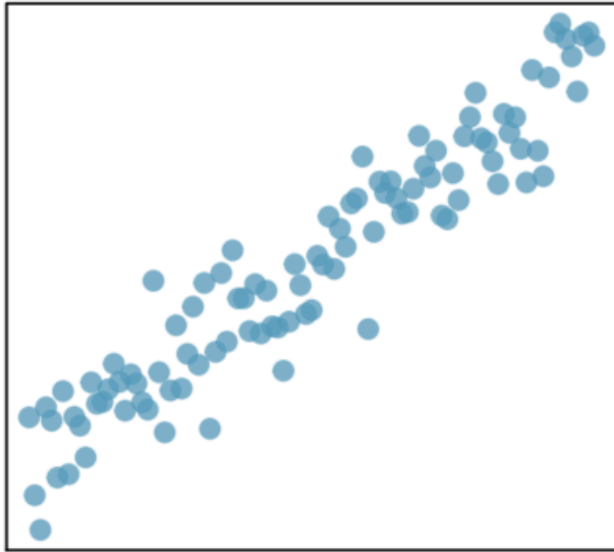
Trend line examples



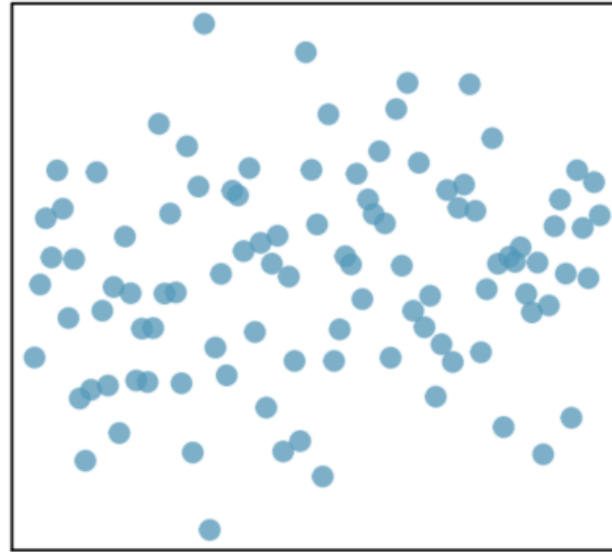
Examples

Describe the scatter plots (1)

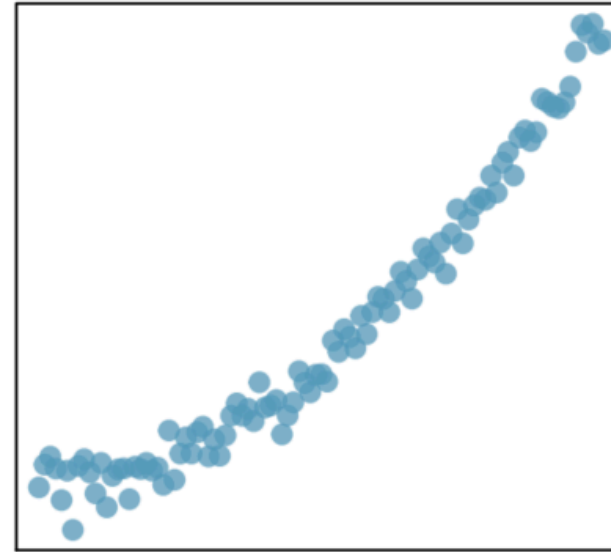
What are the associations between X and Y? (positive/negative, linear/nonlinear)



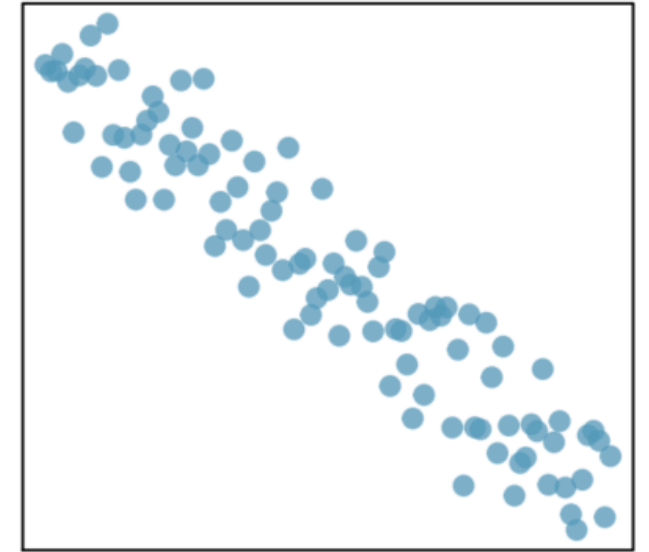
(1)



(2)



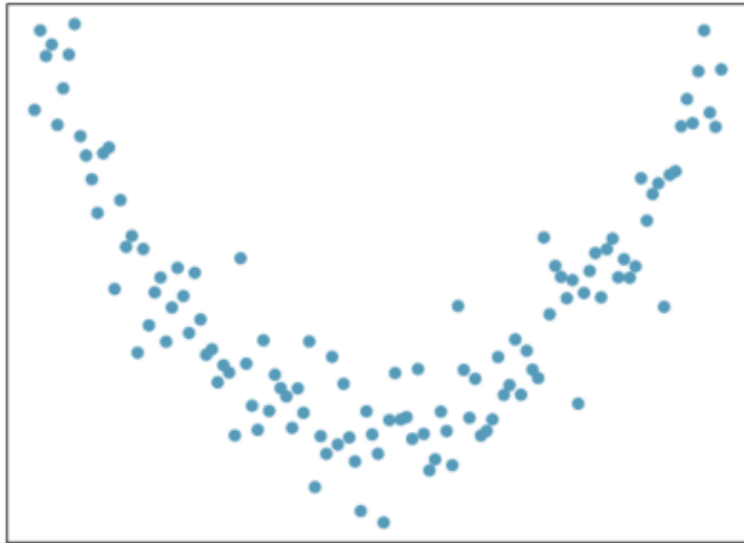
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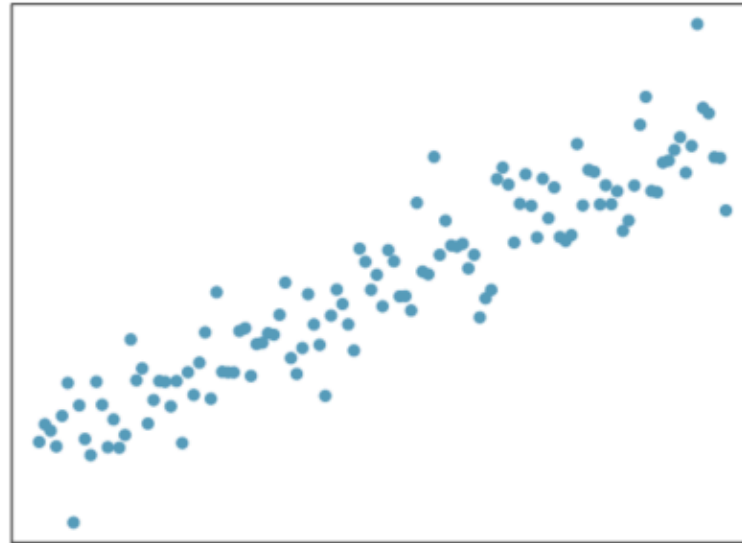
(4)

Describe the scatter plots (2)

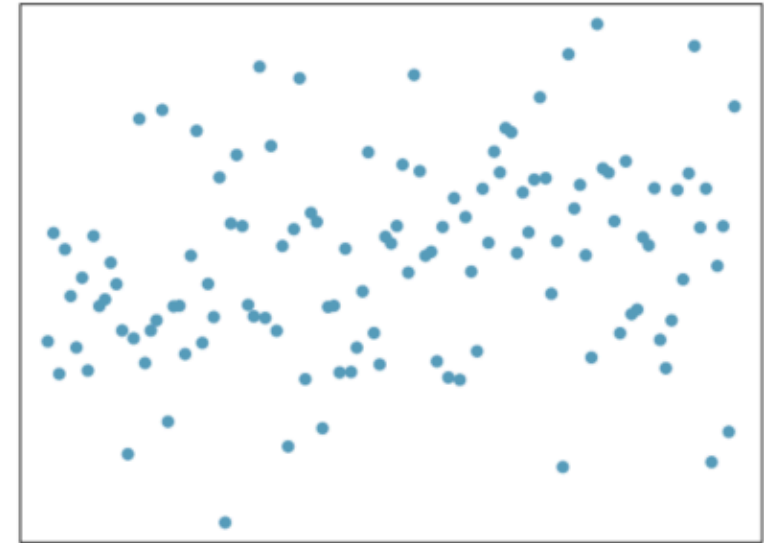
What are the associations between X and Y? (positive/negative, linear/nonlinear)



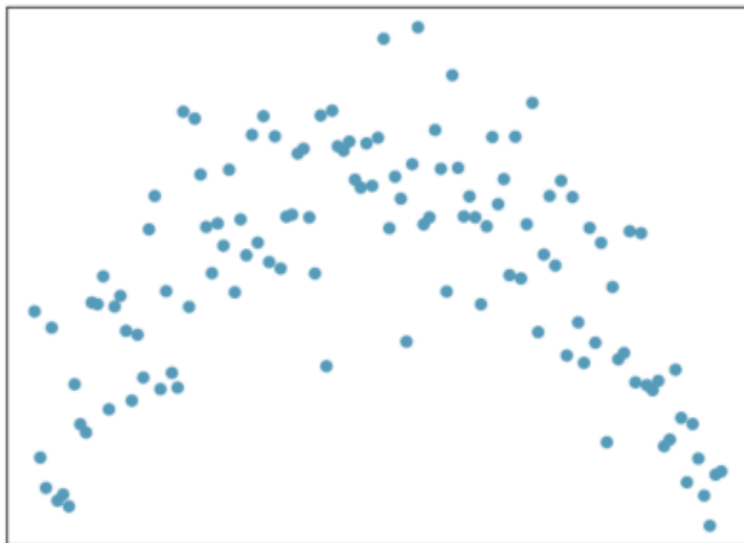
(a)



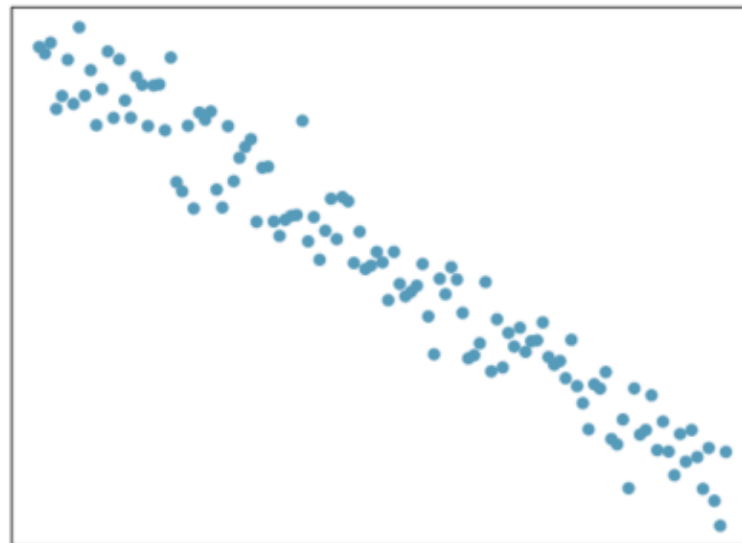
(b)



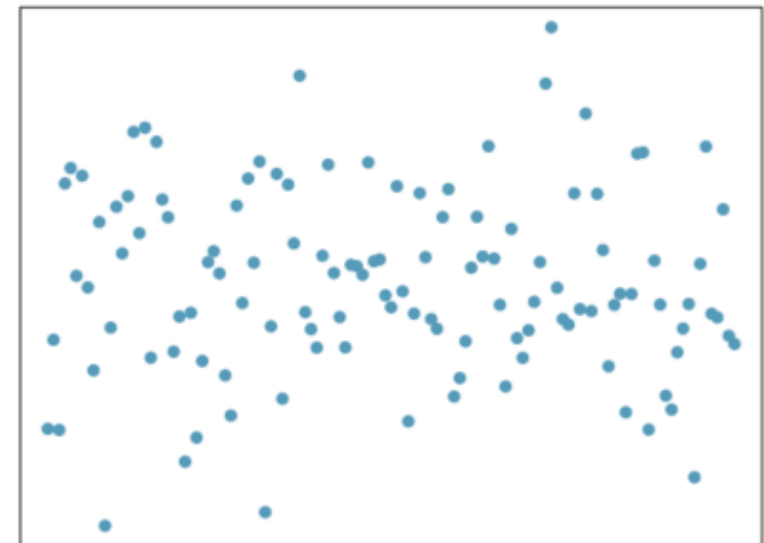
(c)



(d)



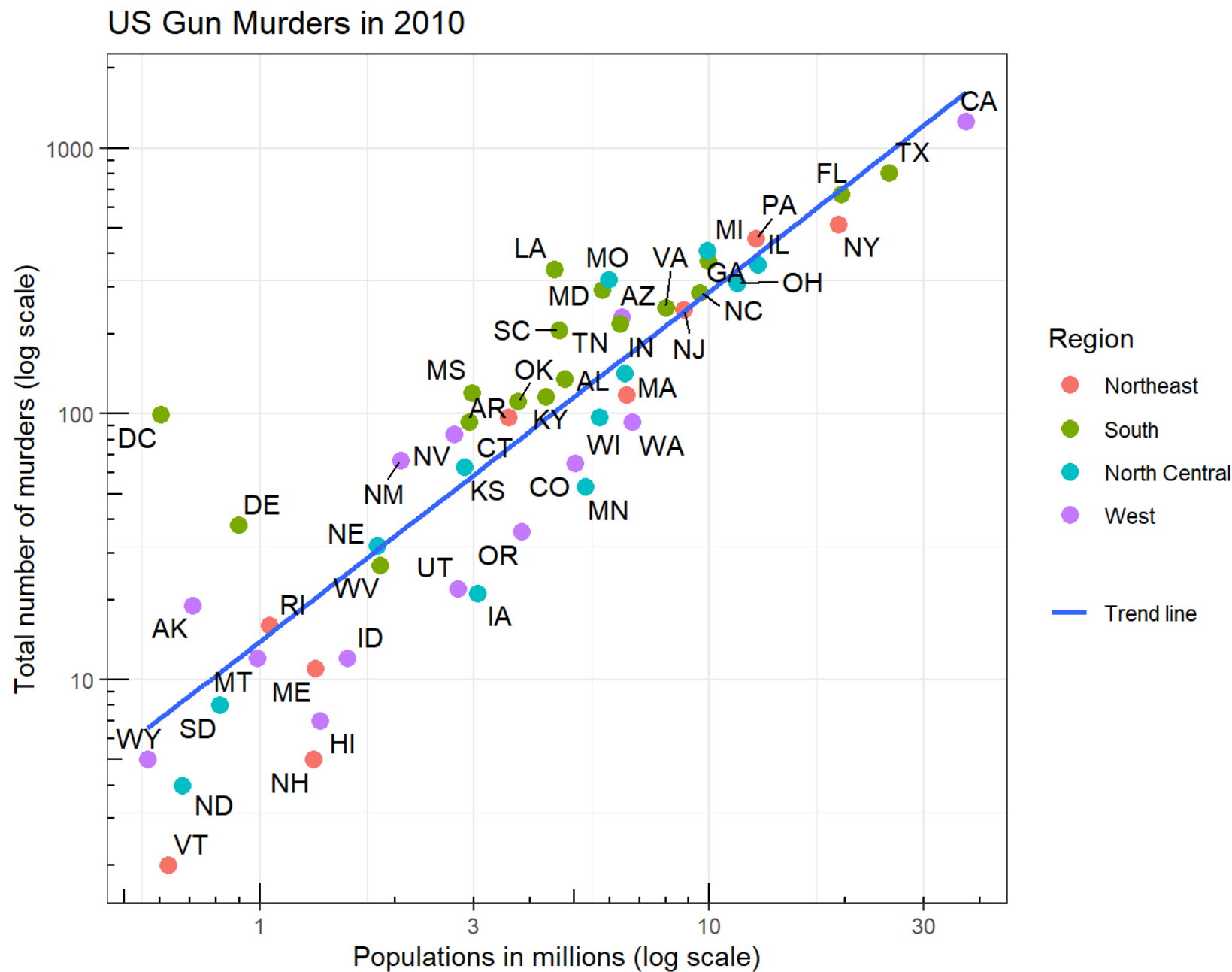
(e)



(f)

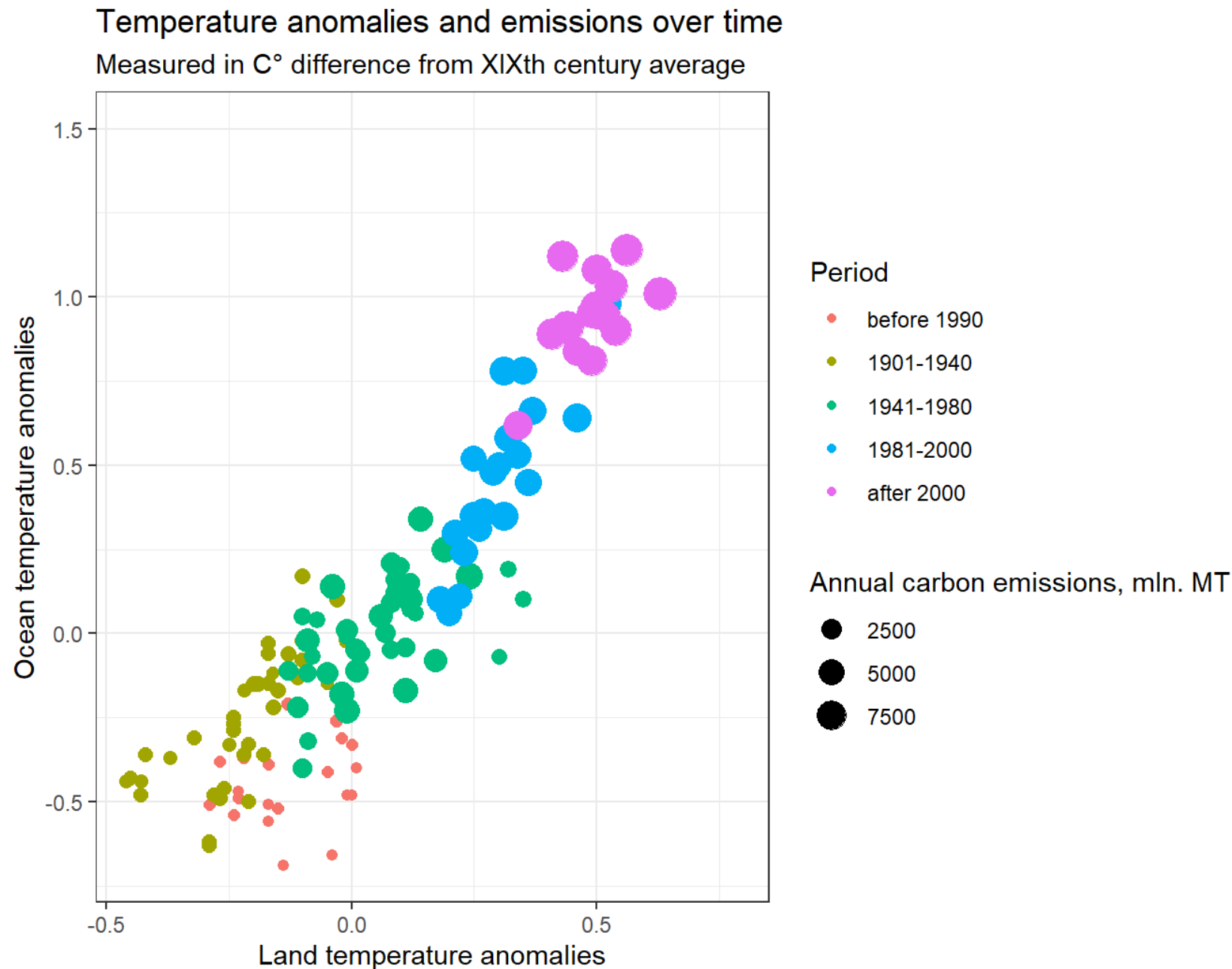
Homework

More examples (1)



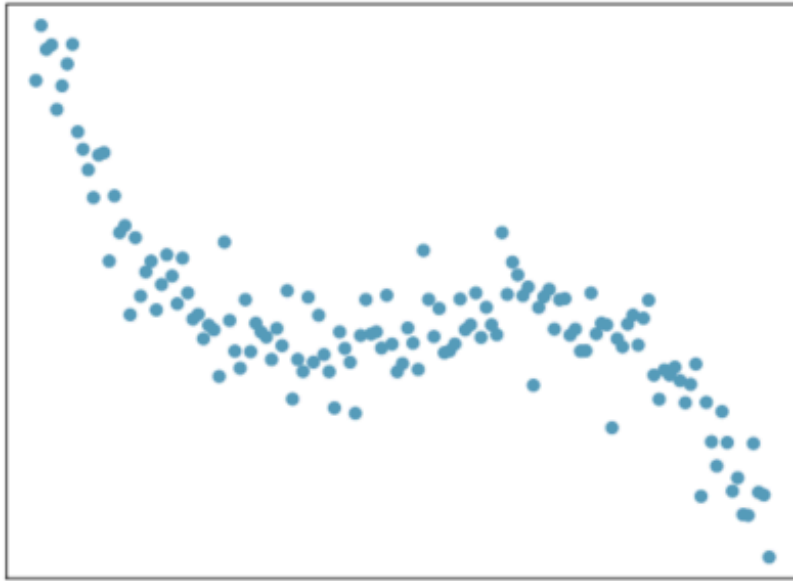
- Describe the scatter plot.
 - What is displayed? What is the plot type?
 - What is on X and Y axis? Are those transformed, how?
 - What points show. What shape/color/size stands for?
- Summarizes the relationship between Y and X.
 - Positive/negative.
 - Linear non-linear.

More examples (2)

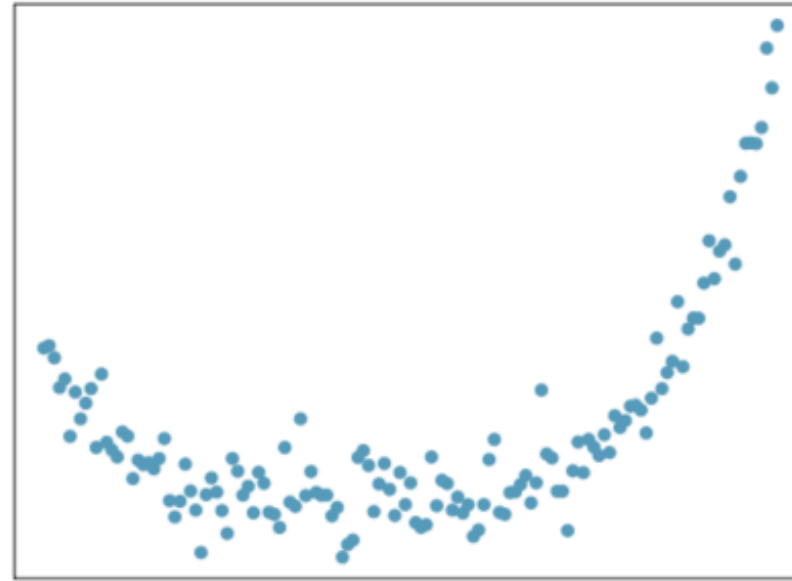


- Describe the scatter plot.
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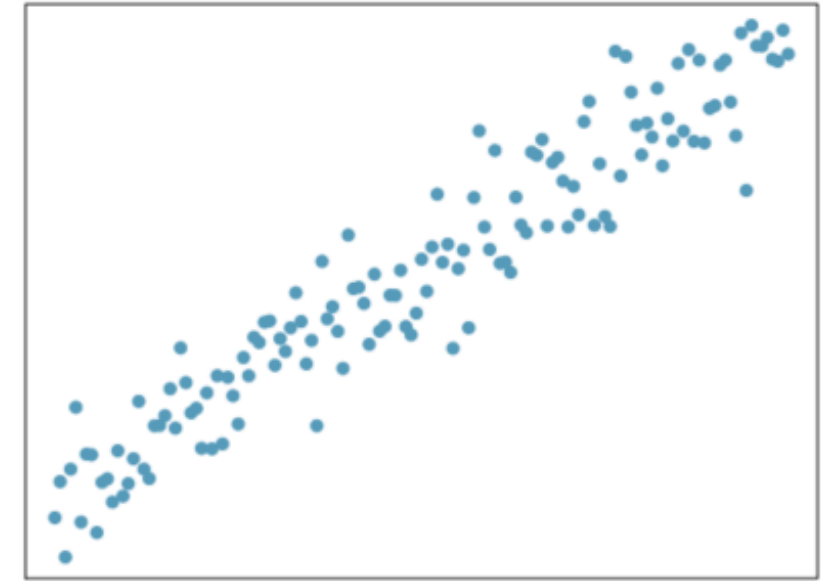
Draw a linear trend line between x and y



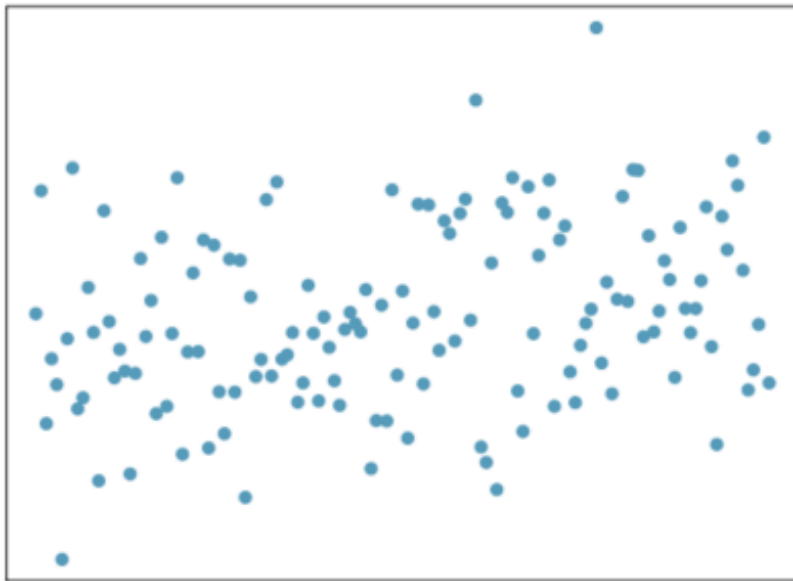
(a)



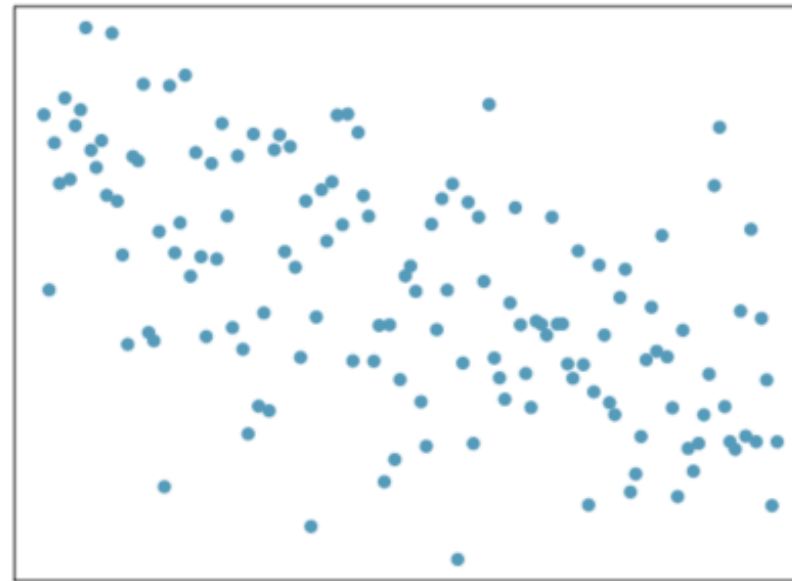
(b)



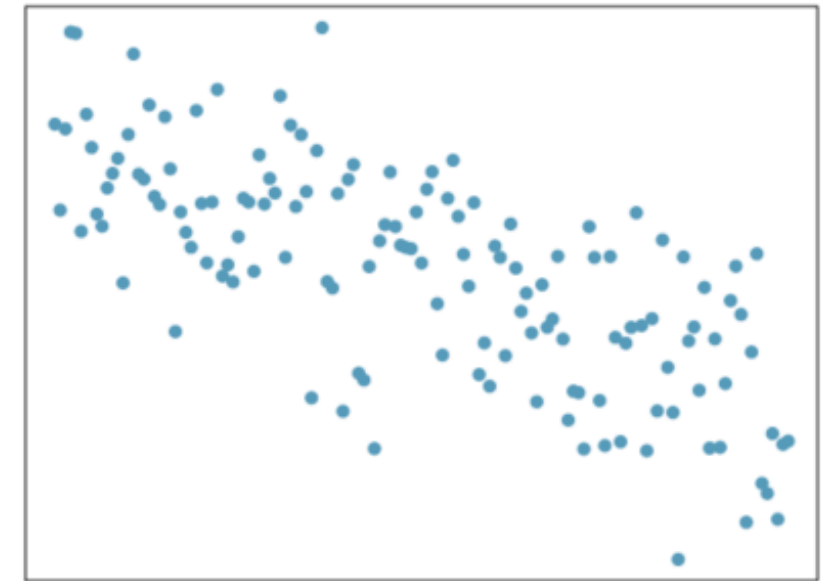
(c)



(d)



(e)



(f)

Correlation (coefficient)

is a measure of a statistical relationship (not necessarily causal) between two variables.

- Tells us how strongly/weakly two variables co-relate with each other.

See:

- Josh Starmer: [Pearson's Correlation, Clearly Explained!](#)
- [Interpretation of the strength of correlation.](#)
- Khan Academy. [Unit: Exploring bivariate numerical data. Correlation coefficients](#)

Methodology

There are many [methods of calculating a correlation coefficient](#):

Pearson's correlation coefficient

Population correlation

$$r_{xy} = \frac{1}{N} \sum_{i=1}^N \frac{x_i - \mu_x}{\sigma_x} \frac{y_i - \mu_y}{\sigma_y}$$

where:

- r_{xy} is the correlation coefficient;
- x_i and y_i are bi-variate observations;
- μ_x and μ_y are the population means;
- σ_x and σ_y are the population standard deviations;

Sample correlation

$$R_{xy} = \frac{1}{n-1} \sum_{i=1}^n \frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y}$$

where:

- R_{xy} is the correlation coefficient;
- $\bar{x}$ and $\bar{y}$ are the sample means;
- s_x and s_y are the sample standard deviations;

Interpretation

Correlation coefficient represents **ONLY LINEAR** relationship between two variables

It describe the strength of a relationship and its direction

Strength of a relationship by [J. Cohen \(1988\)](#)

- $r < 0.1$ Very small / weak
- $0.1 \leq r < 0.3$ Small
- $0.3 \leq r < 0.5$ Moderate
- $r \geq 0.5$ Large / strong

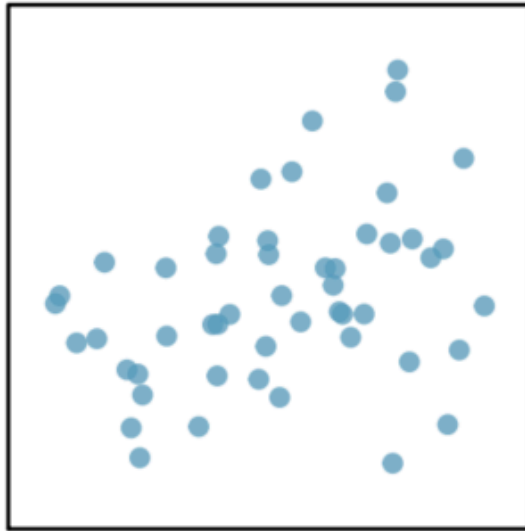
Positive correlation: $r > 0$

- x and y relate proportionally.
- An **increase** in x (y) is associated with an **increase** in y (x).

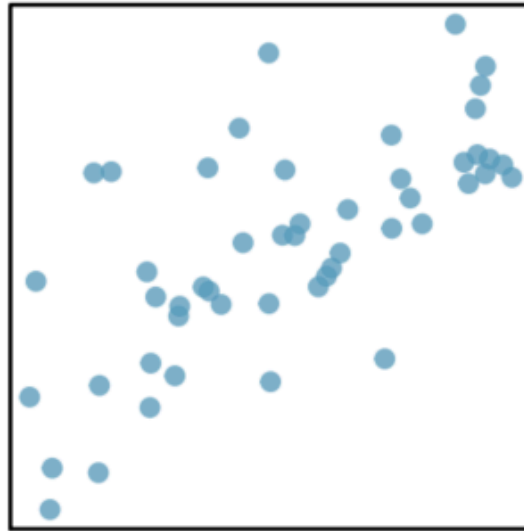
Negative correlation: $r < 0$

- x and y relate inversely;
- An **increase** in x (y) is associated with a **decrease** in y (x).

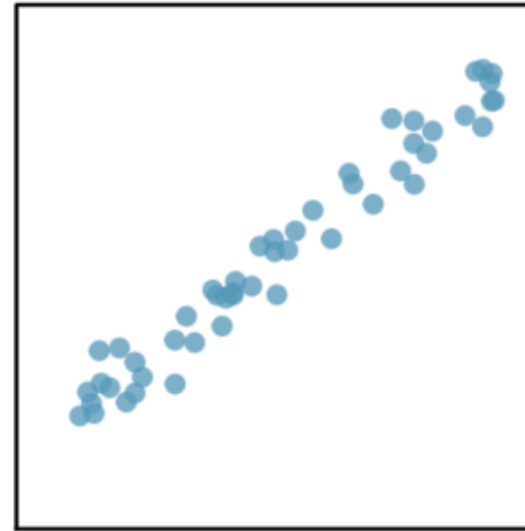
Correlation examples (1)



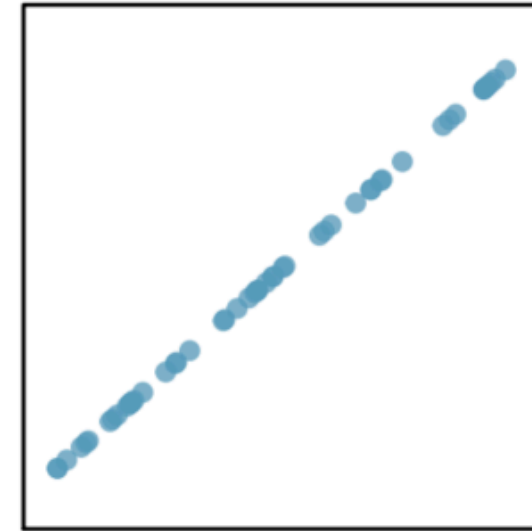
$R = 0.33$



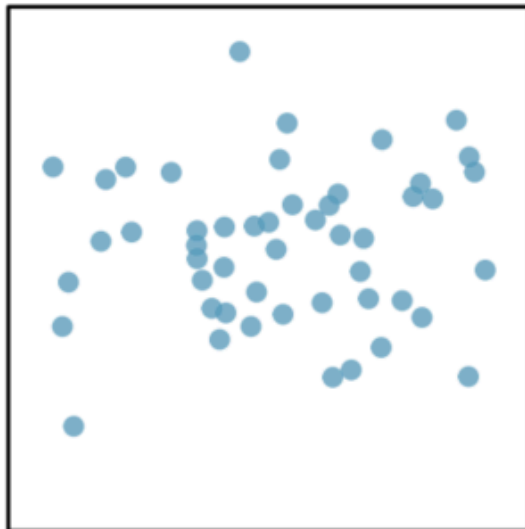
$R = 0.69$



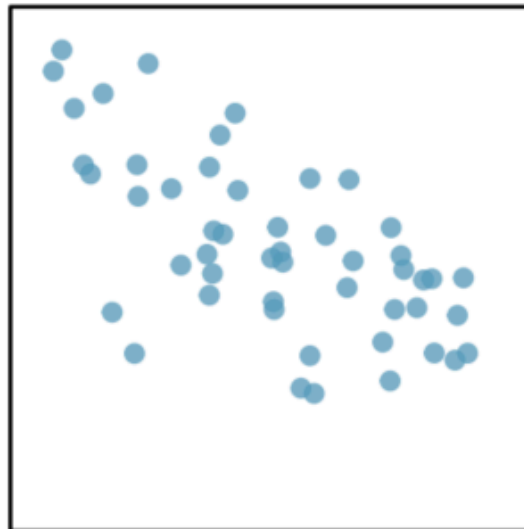
$R = 0.98$



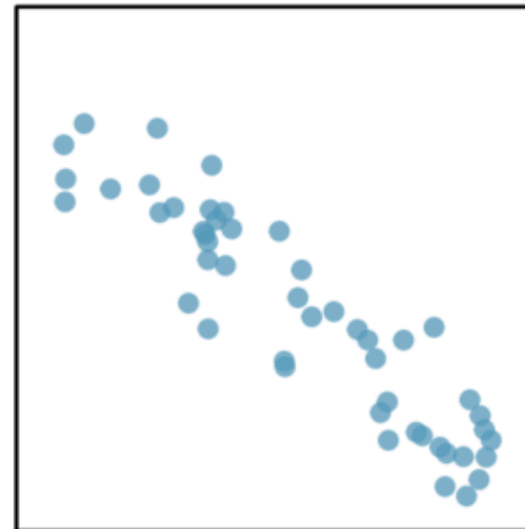
$R = 1.00$



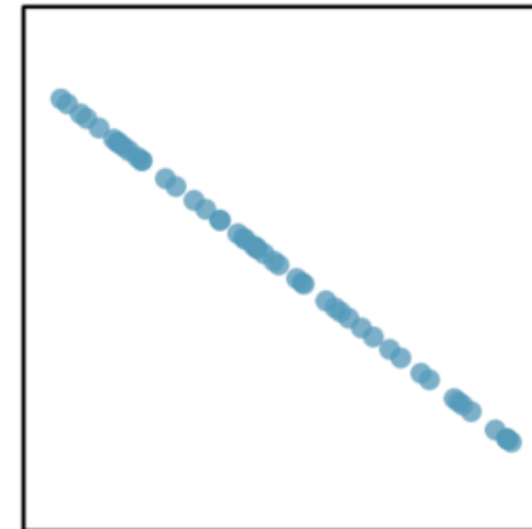
$R = 0.08$



$R = -0.64$

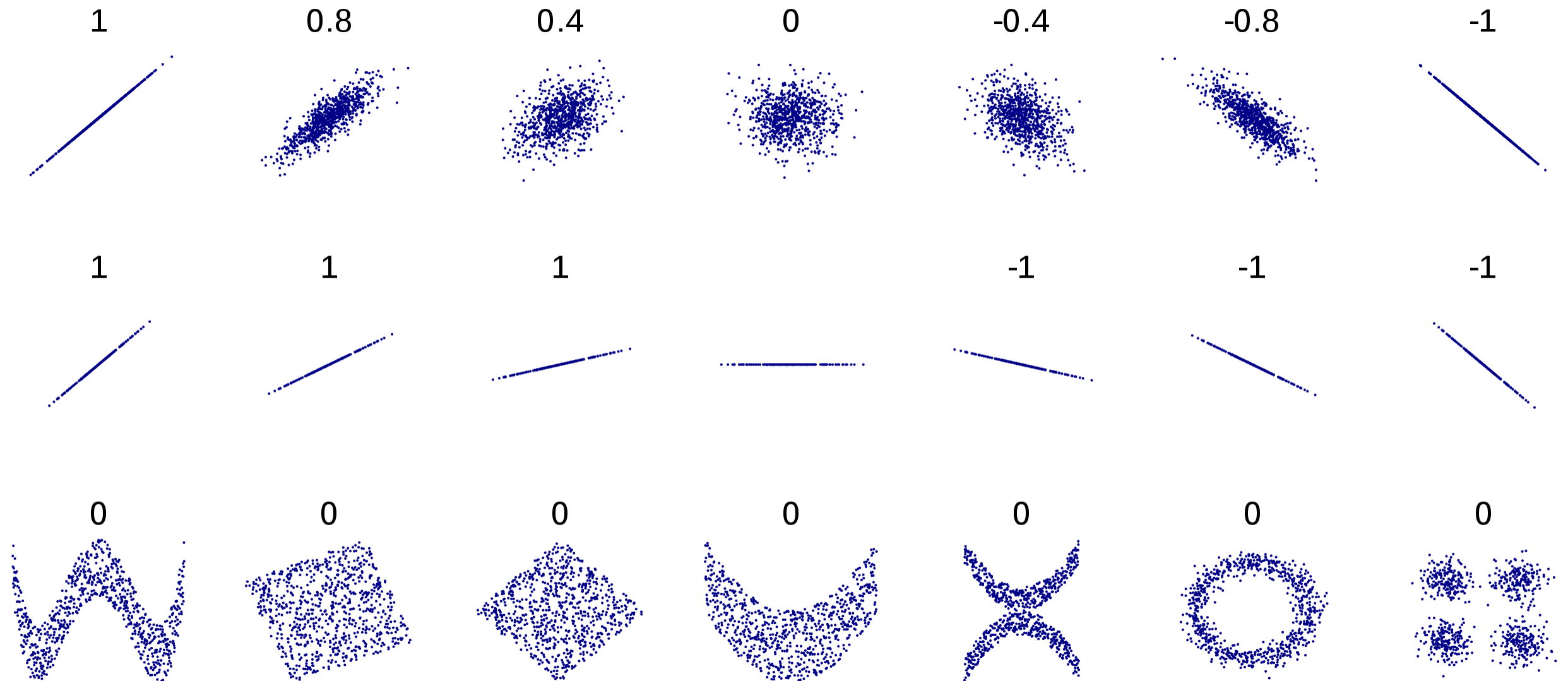


$R = -0.92$



$R = -1.00$

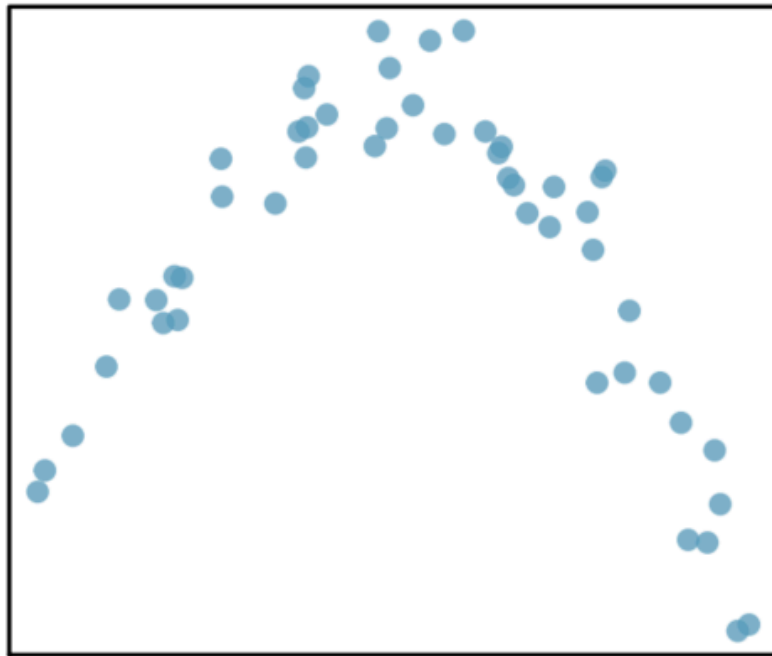
Correlation examples (2)



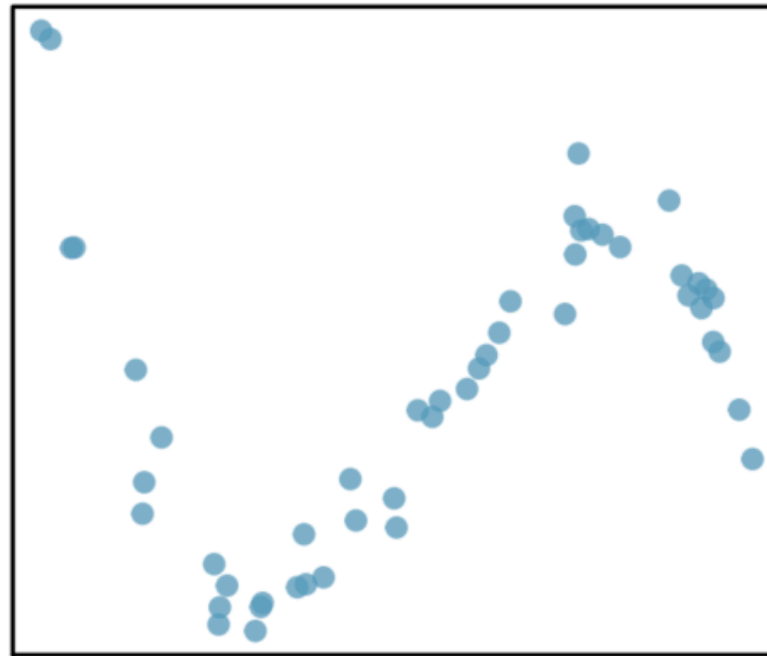
Correlation pitfalls: Non-linearity

Correlation cannot be used with the non-linear data.

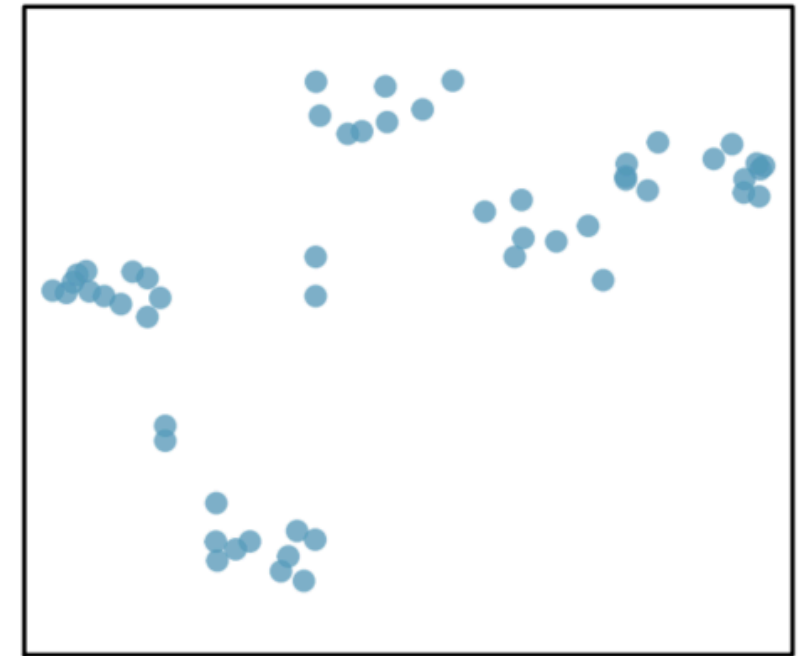
- Non linear data yields a correlation coefficient, but such coefficient does not imply any linear relationship.



$R = -0.23$

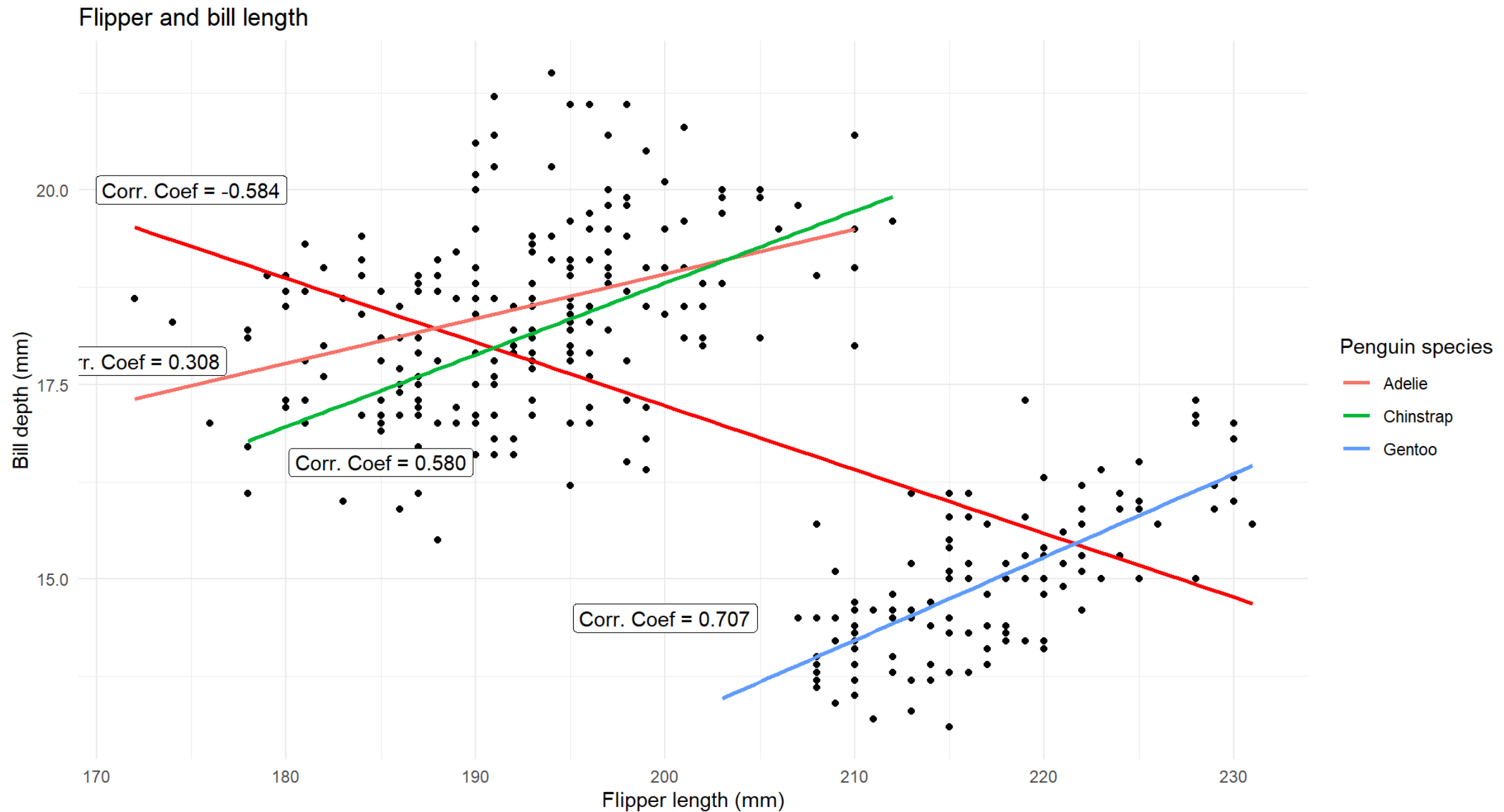


$R = 0.31$



$R = 0.50$

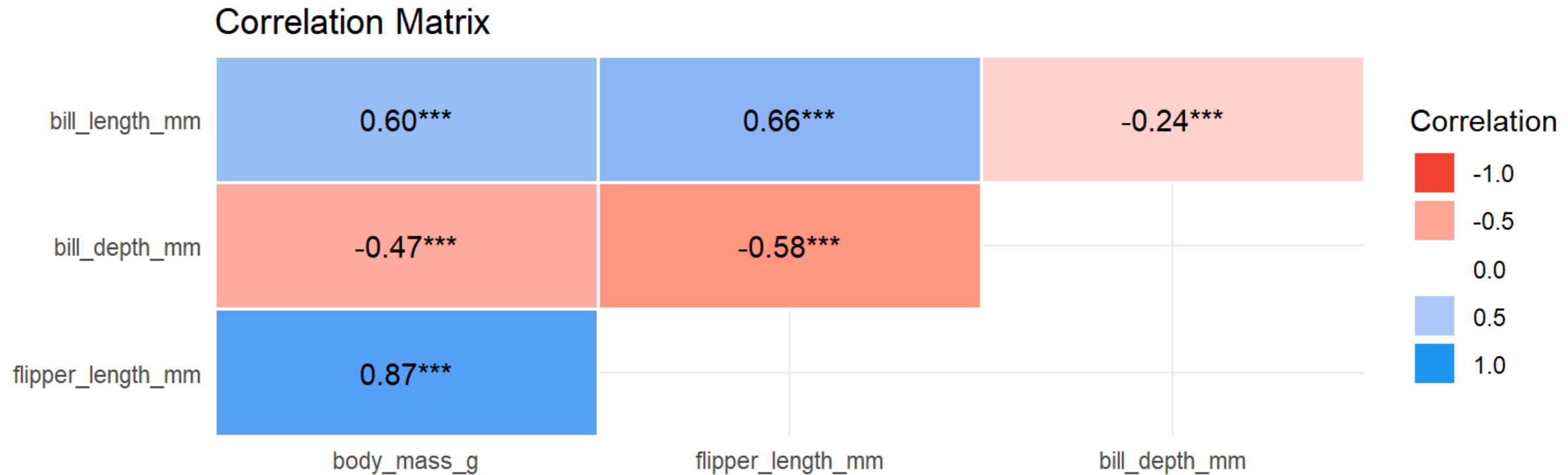
Correlation pitfalls



Correlation matrix (1): Penguins

Parameter1	Parameter2	r
bill_length_mm	bill_depth_mm	-0.2350529
bill_length_mm	flipper_length_mm	0.6561813
bill_length_mm	body_mass_g	0.5951098
bill_depth_mm	flipper_length_mm	-0.5838512
bill_depth_mm	body_mass_g	-0.4719156
flipper_length_mm	body_mass_g	0.8712018

Parameter	body_mass_g	flipper_length_mm	bill_depth_mm
bill_length_mm	0.5951098	0.6561813	-0.2350529
bill_depth_mm	-0.4719156	-0.5838512	
flipper_length_mm	0.8712018		

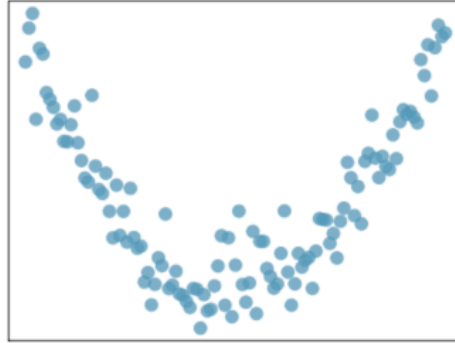


Homework

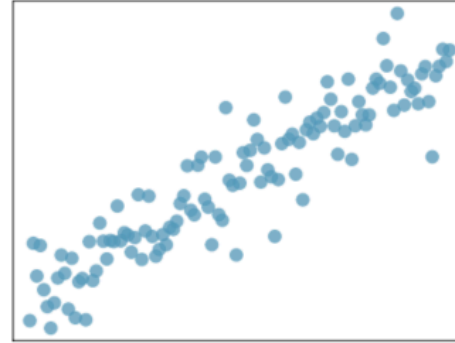
Correlation

Match a correlation coefficient and a plot

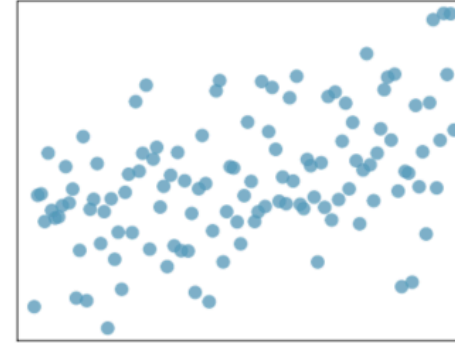
- (a) $R = -0.7$
- (b) $R = 0.45$
- (c) $R = 0.06$
- (d) $R = 0.92$



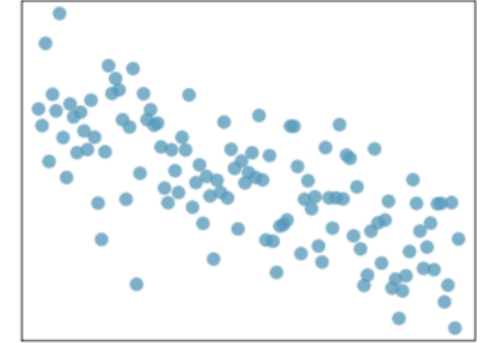
(1)



(2)



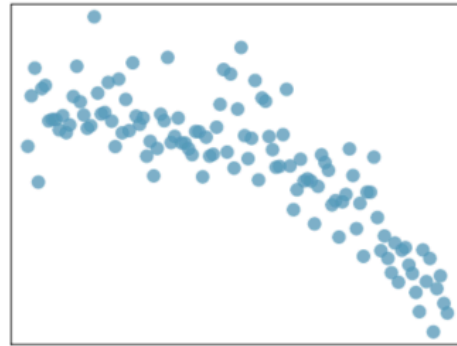
(3)



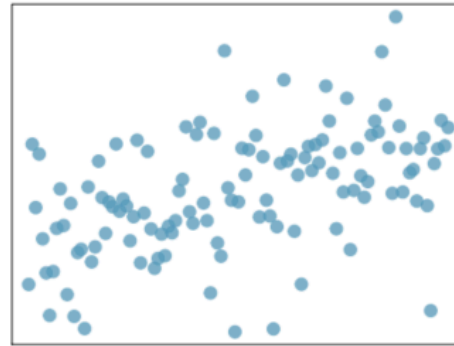
(4)

Match a correlation coefficient and a plot

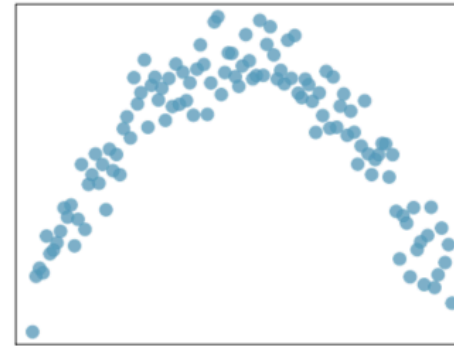
- (a) $R = 0.49$
 (b) $R = -0.48$
 (c) $R = -0.03$
 (d) $R = -0.85$



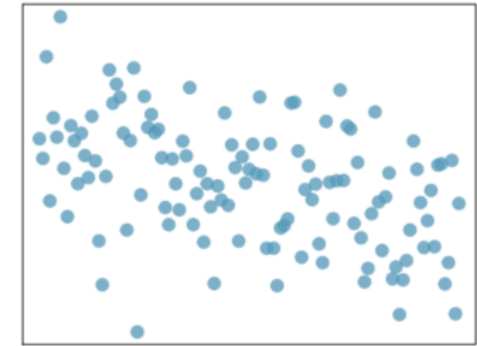
(1)



(2)

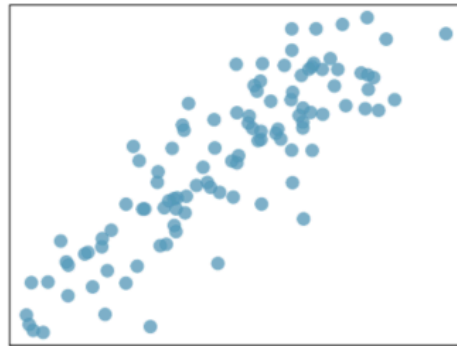


(3)

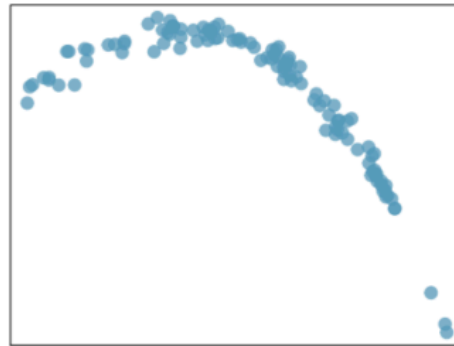


(4)

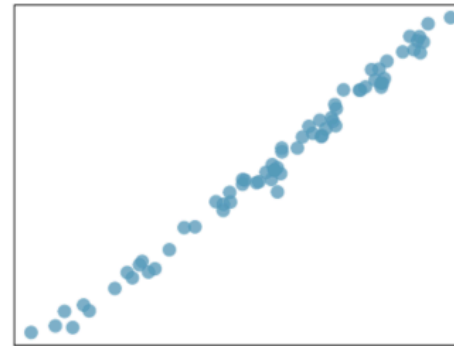
- (a) $r = -0.72$
 (b) $r = 0.07$
 (c) $r = 0.86$
 (d) $r = 0.99$



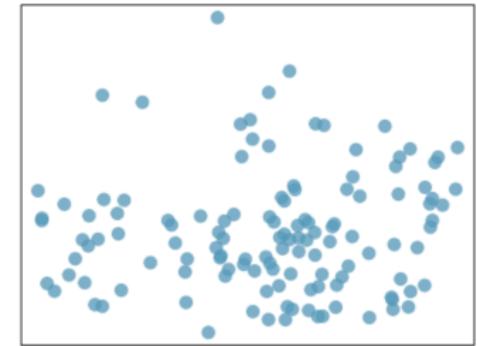
(1)



(2)

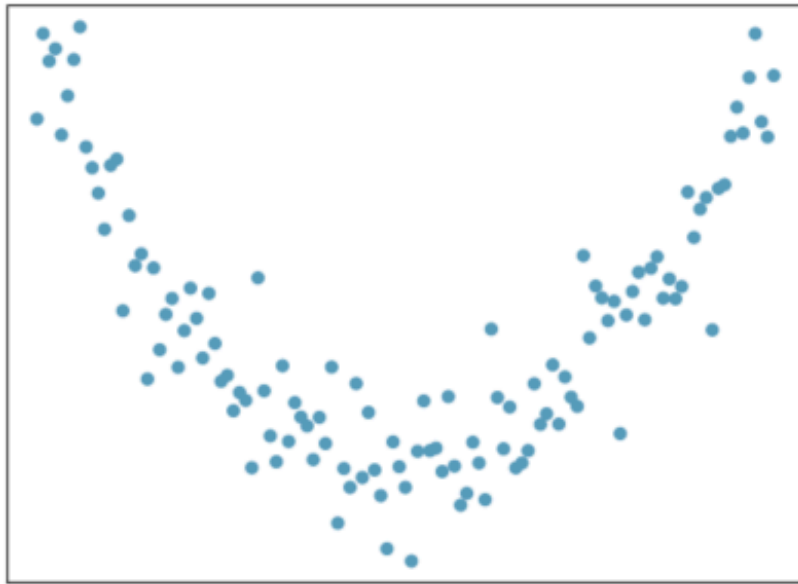


(3)

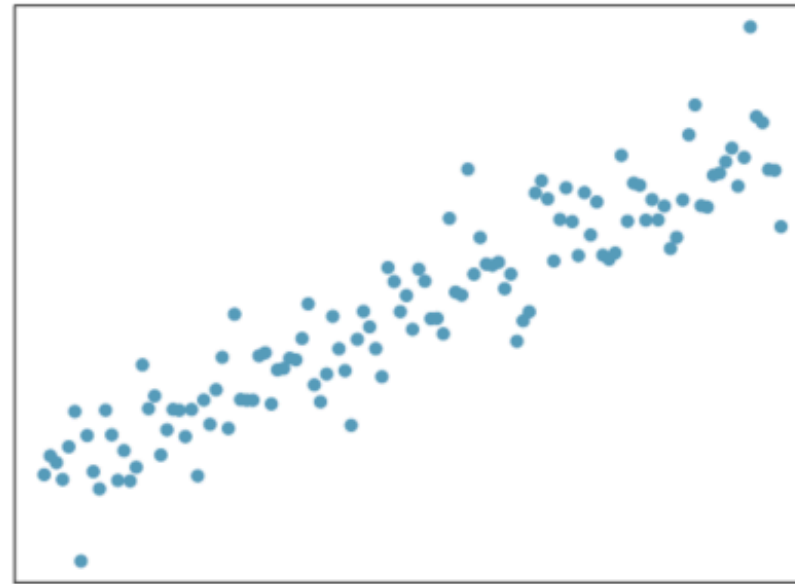


(4)

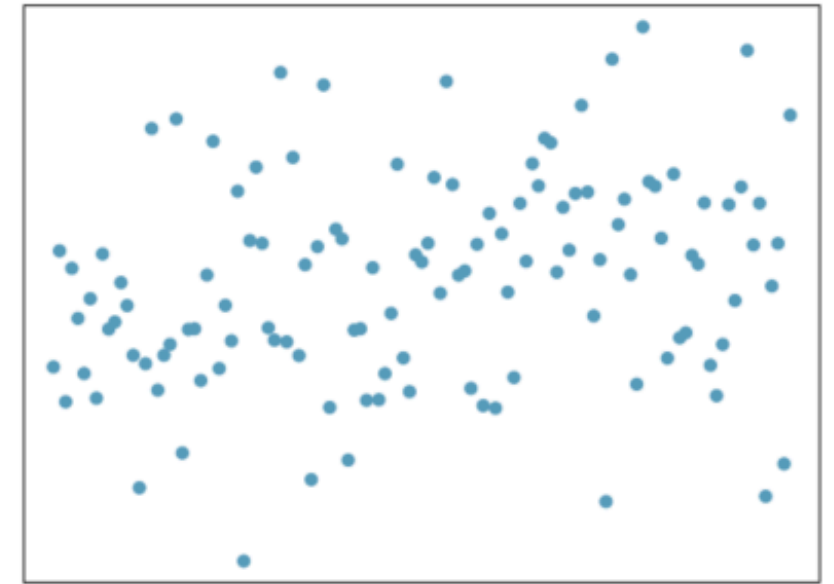
Identify the relationship and its strength (1)



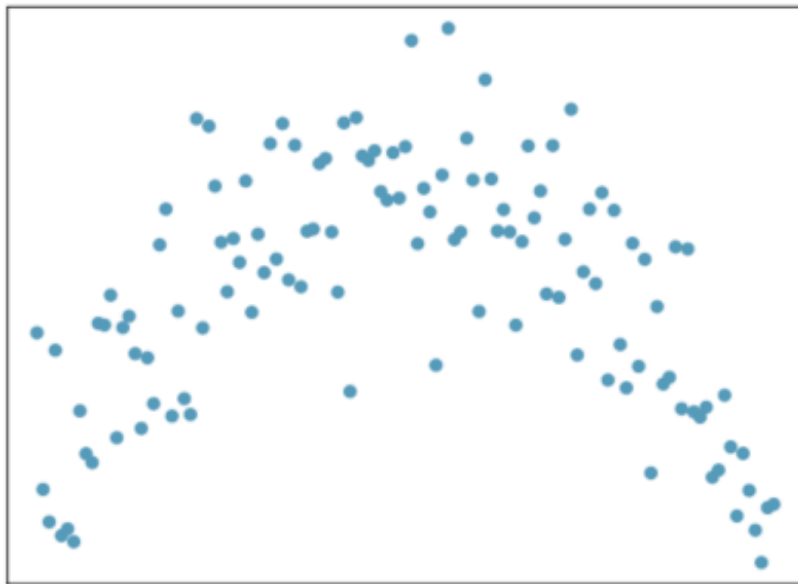
(a)



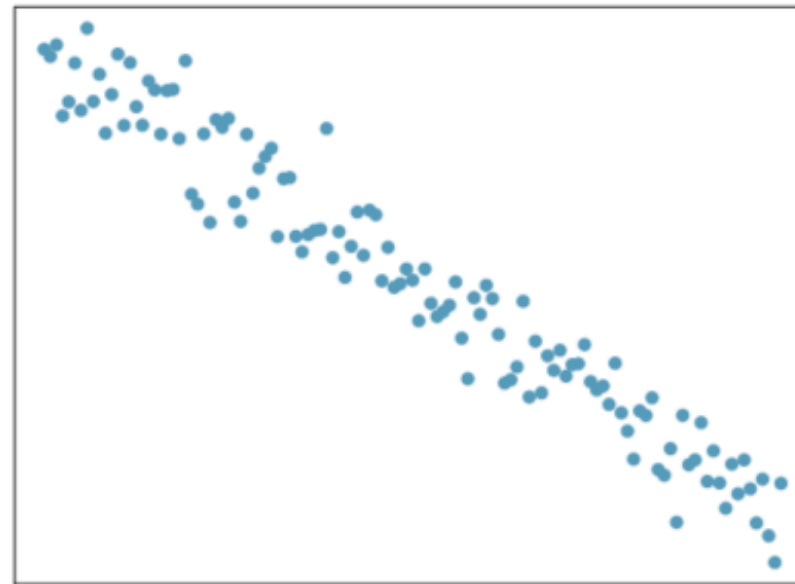
(b)



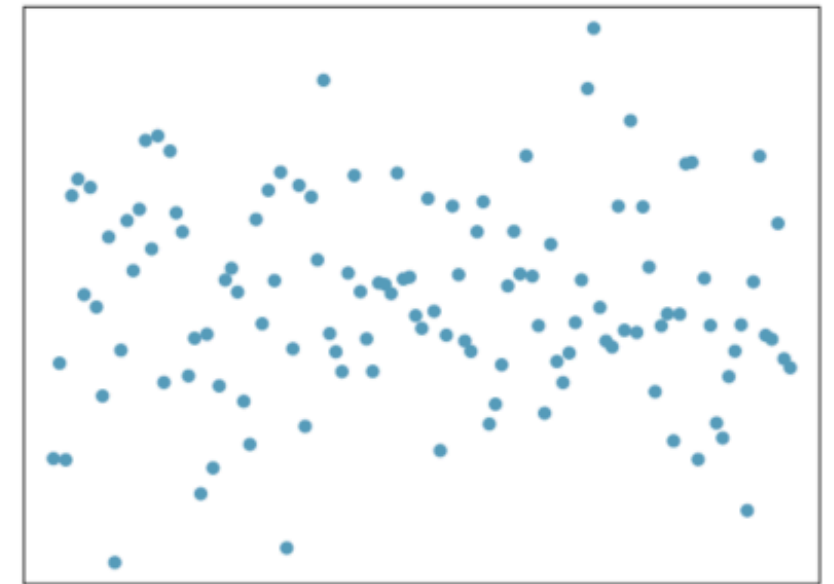
(c)



(d)

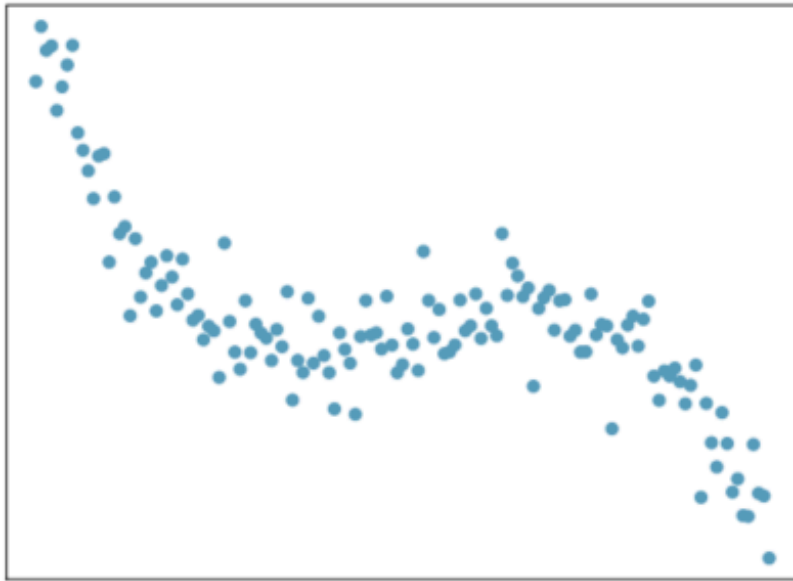


(e)

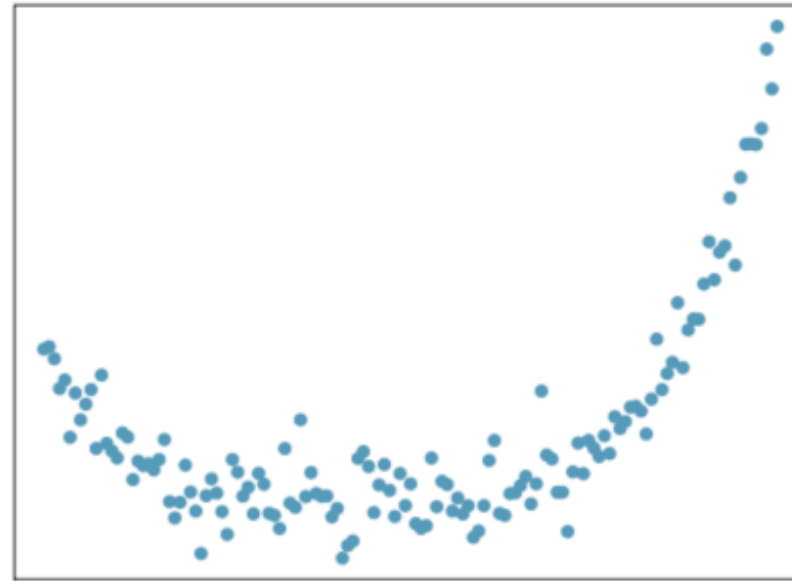


(f)

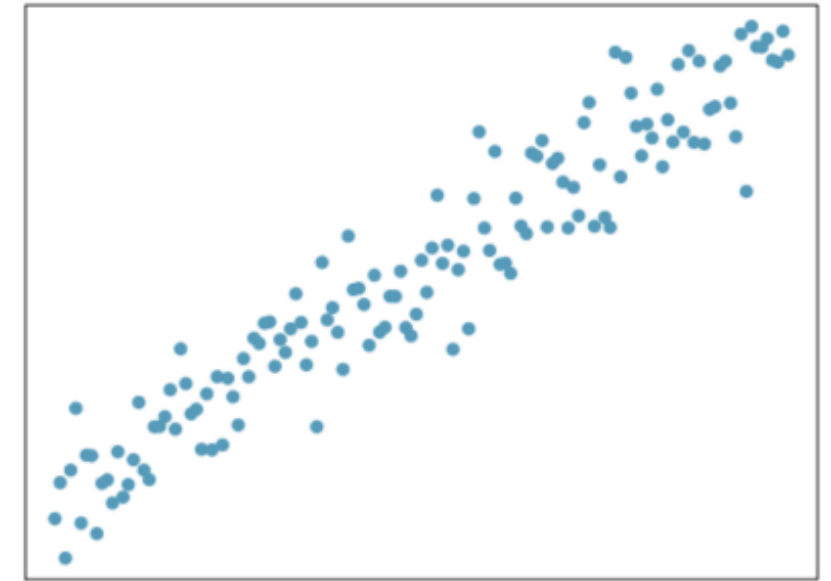
Identify the relationship and its strength (2)



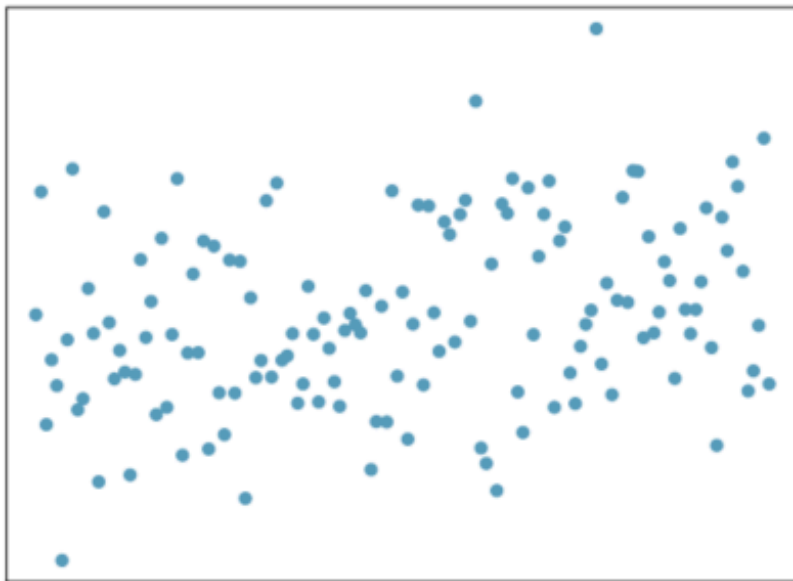
(a)



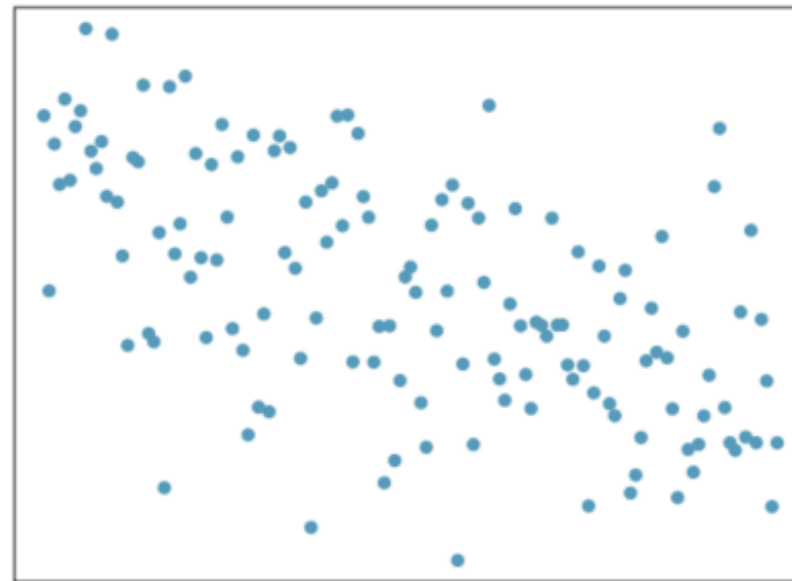
(b)



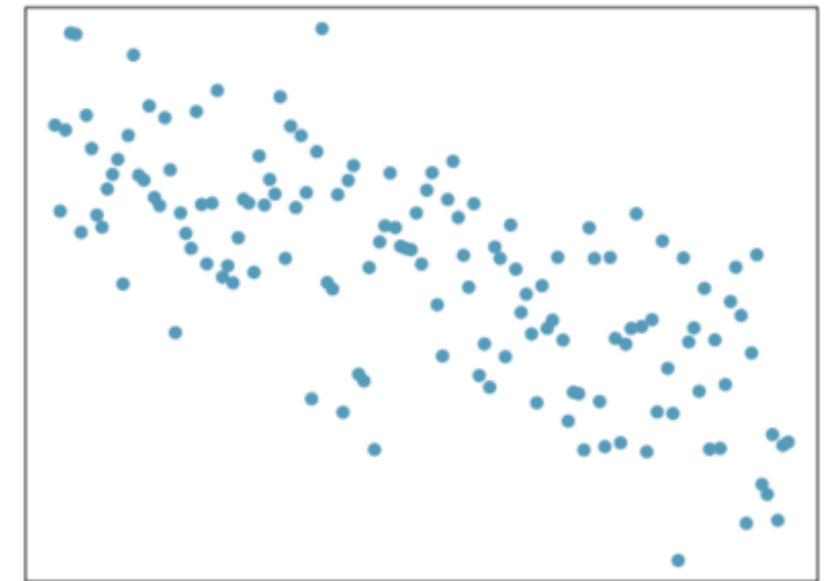
(c)



(d)



(e)



(f)

Spurious correlation (link)

Explore at home on your own:

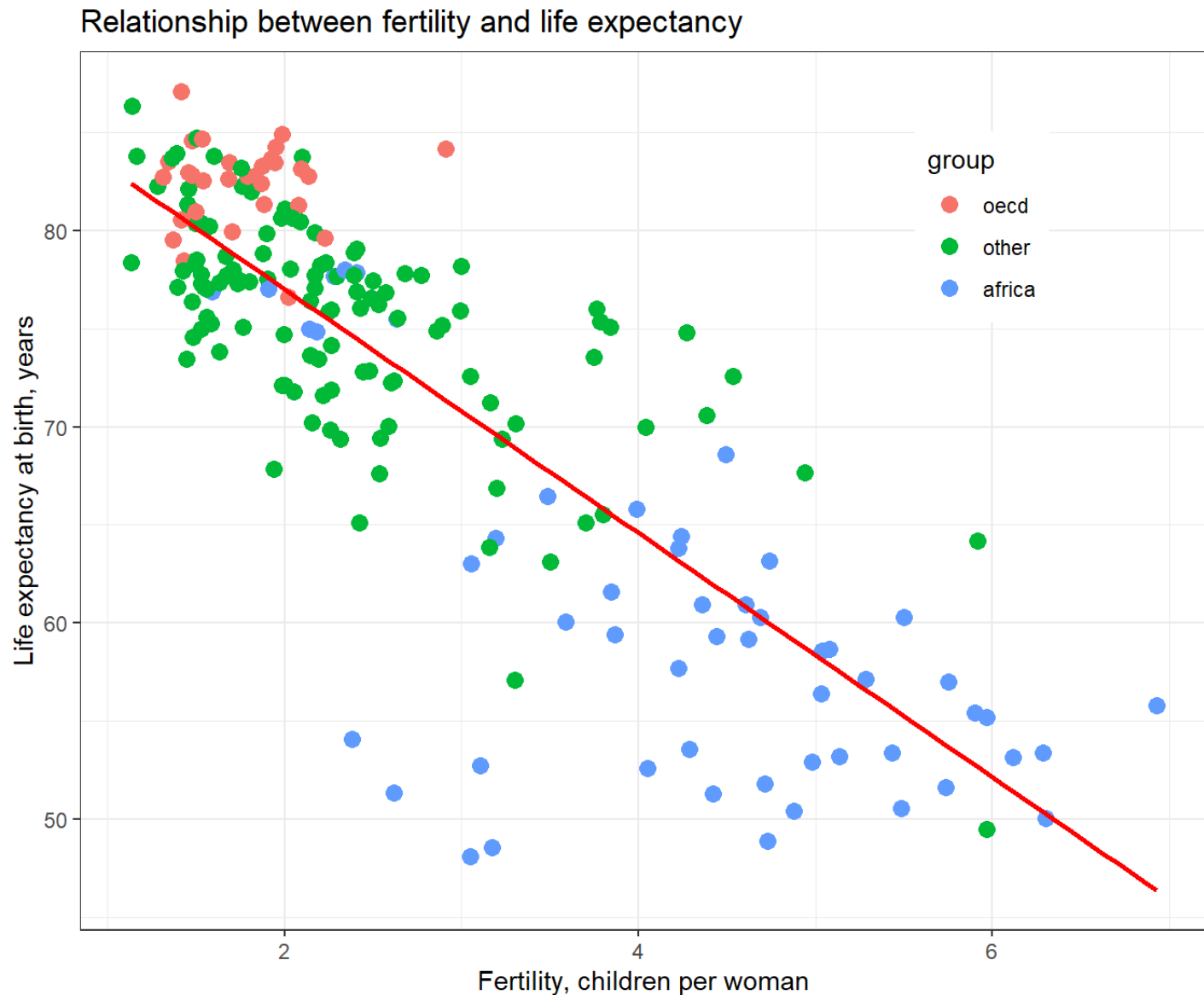
- <https://www.tylervigen.com/spurious-correlations>
- Define what is “spurious correlation”?

Correlation and Causation

If x correlates with y , does it mean that x causes y ?

- Correlation does not imply causation!

Do we live longer because of the low birth rate?



Data source: Weisberg, S. (2014). Applied Linear Regression

The correlation between Fertility and life expectancy is -0.824.

- What does this mean?
- Can we imply that giving fewer births will increase life expectancy?
- What does the linear trend suggests?
- Does being from a different country groups causes this trend?

No... None of it implies causality.

To establish a causal relationship

- We need to establish a “Ceteris Paribus”!
- We need to make sure that “other things are equal” between the countries.
- For example:
 - All countries consist of identical people with identical socioeconomic characteristics.
 - The only differences are in their fertility rates and (perhaps) life expectancy.

Homework

To better understand how data can be used to communicate a complex story, check out the following videos and exercises.

New insights on the HIV pandemic

The purpose of the following short video is to stimulate debate on the **covariates, and causes** of the HIV pandemic worldwide.

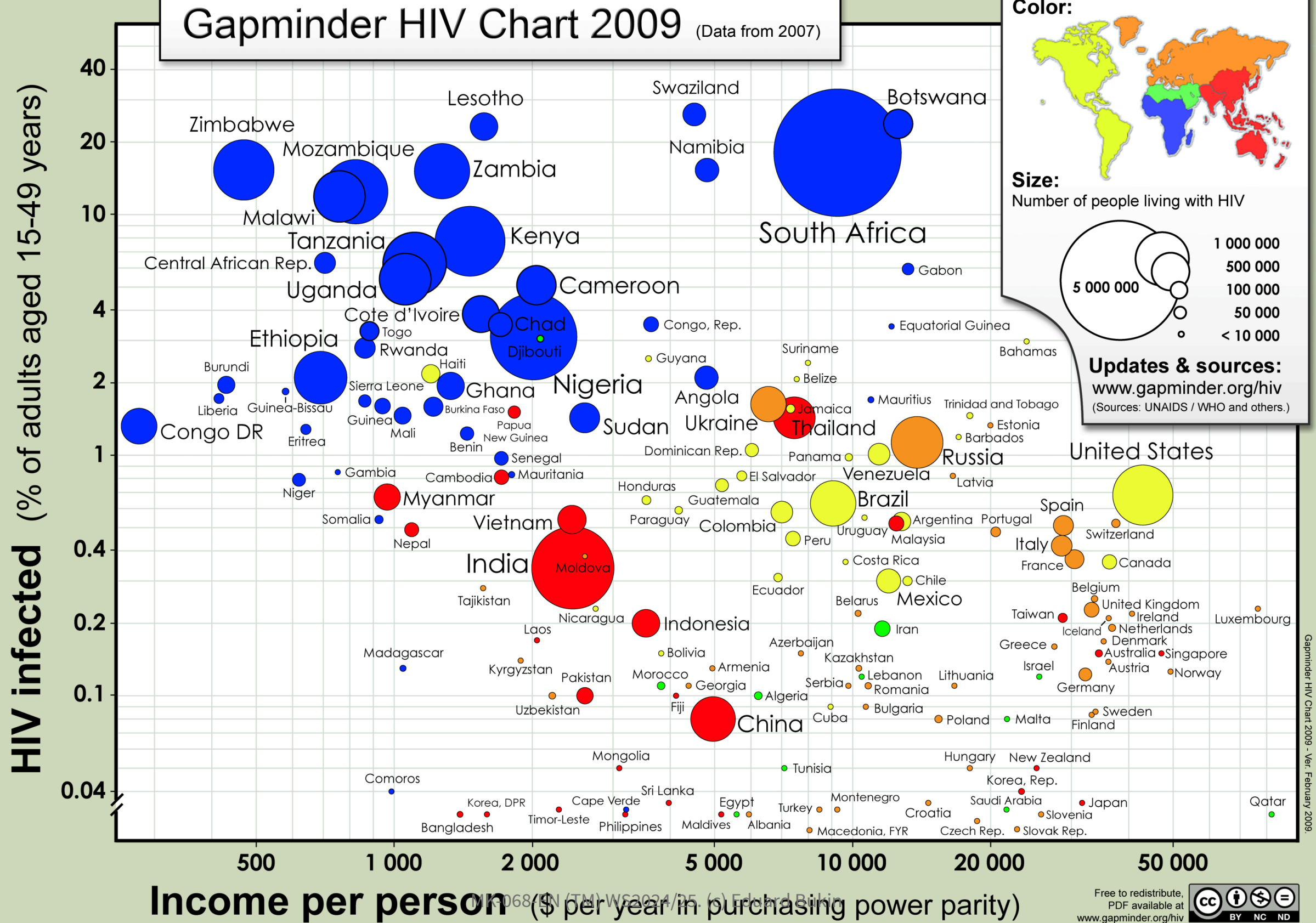
After watching it, discuss:

- Facts of the HIV/AIDS pandemic;
- Country-specific peculiarities that correlate with the pandemic.
- What causes the pandemic to spread at so different rates across the countries?

Video source: [Insights on HIV, in stunning data visuals](#)

Insights on HIV, in stunning data visuals





Key Messages:

- No simple answer to the complex question.
- There are no two types of countries one with and another without the HIV pandemic.

To see more, visit:

- gapminder.org
- [Ted talks of Hans Rosling](#)

At home: Explore the best use of a scatter plots with **Gapminder**

Play with data using gap minder:

- <https://www.gapminder.org/tools/>

Watch some Hans Rosling's Ted Talks:

- https://www.youtube.com/playlist?list=PLSrgd_ElrrXegdtDHPmUzrzLpwZ2dSYKg

Explore the gap-minder foundation channel

- <https://www.youtube.com/channel/UCMbmqUMqzerMSM48pGYMYsg>

Example: Rice yields and fertilizers application

Research question:

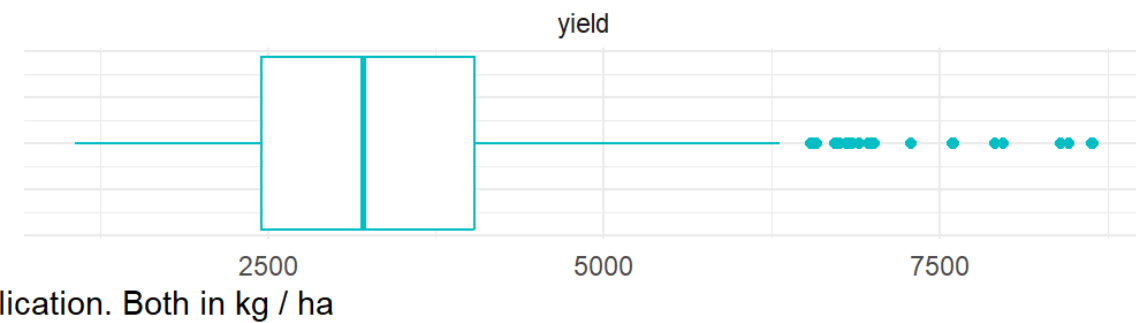
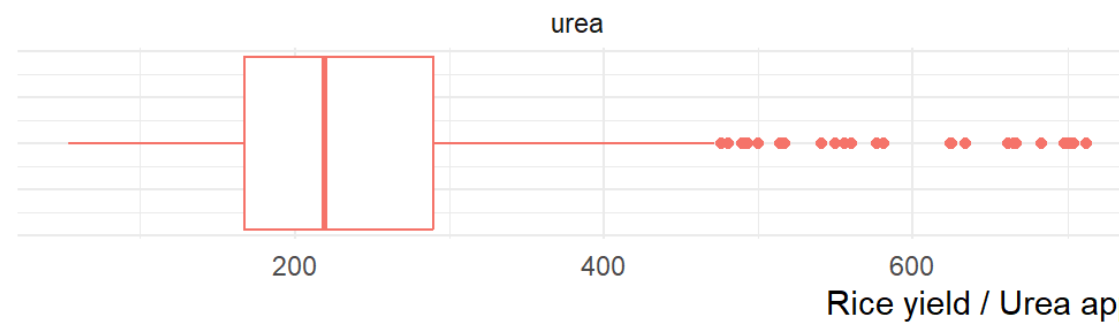
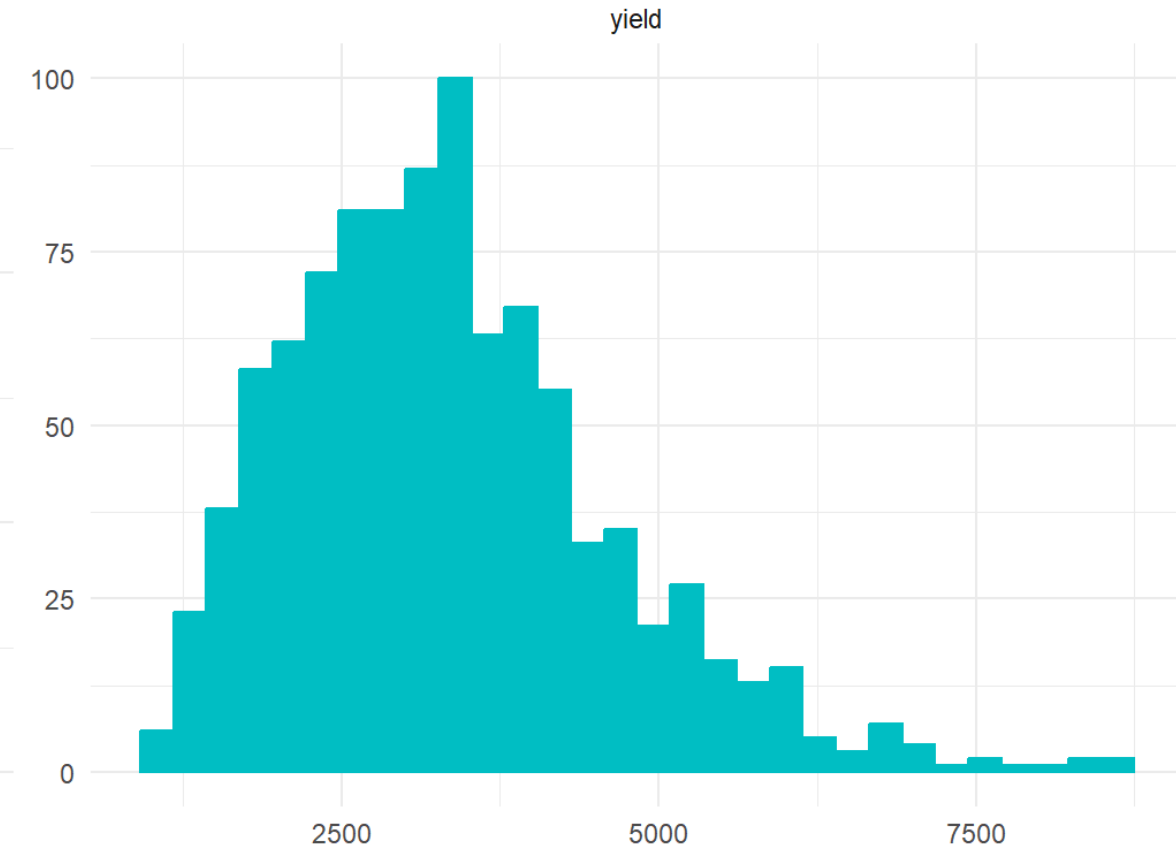
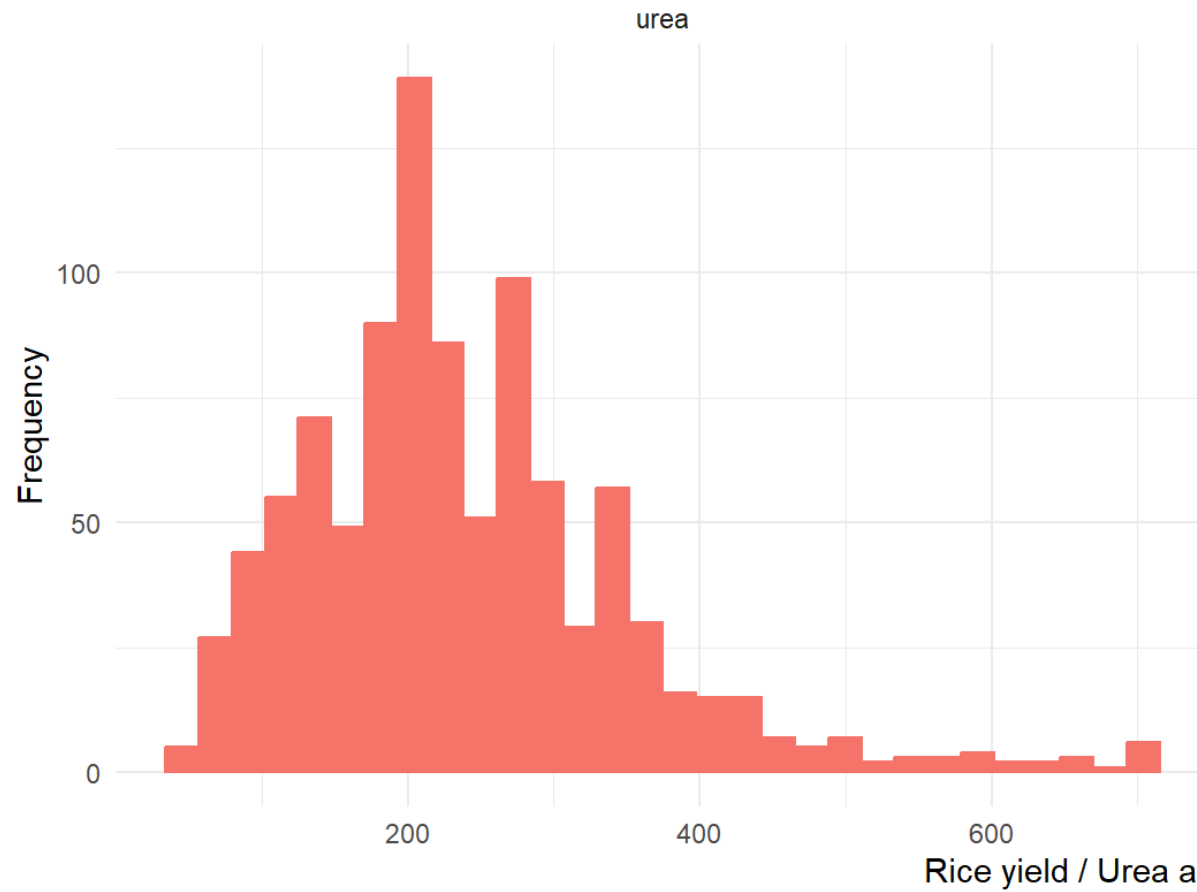
- How does the urea application affect the yields of rice?

The sample of 171 rice farms in Indonesia is followed over 5 years.

- Each observation represents an individual farm that produced rice over one year.
- Two variables:
 - Rice yields in kg / ha
 - Urea ($CO(NH_2)_2$) application in kg / ha

Exploratory analysis

Univariate analysis



Descriptive Statistics

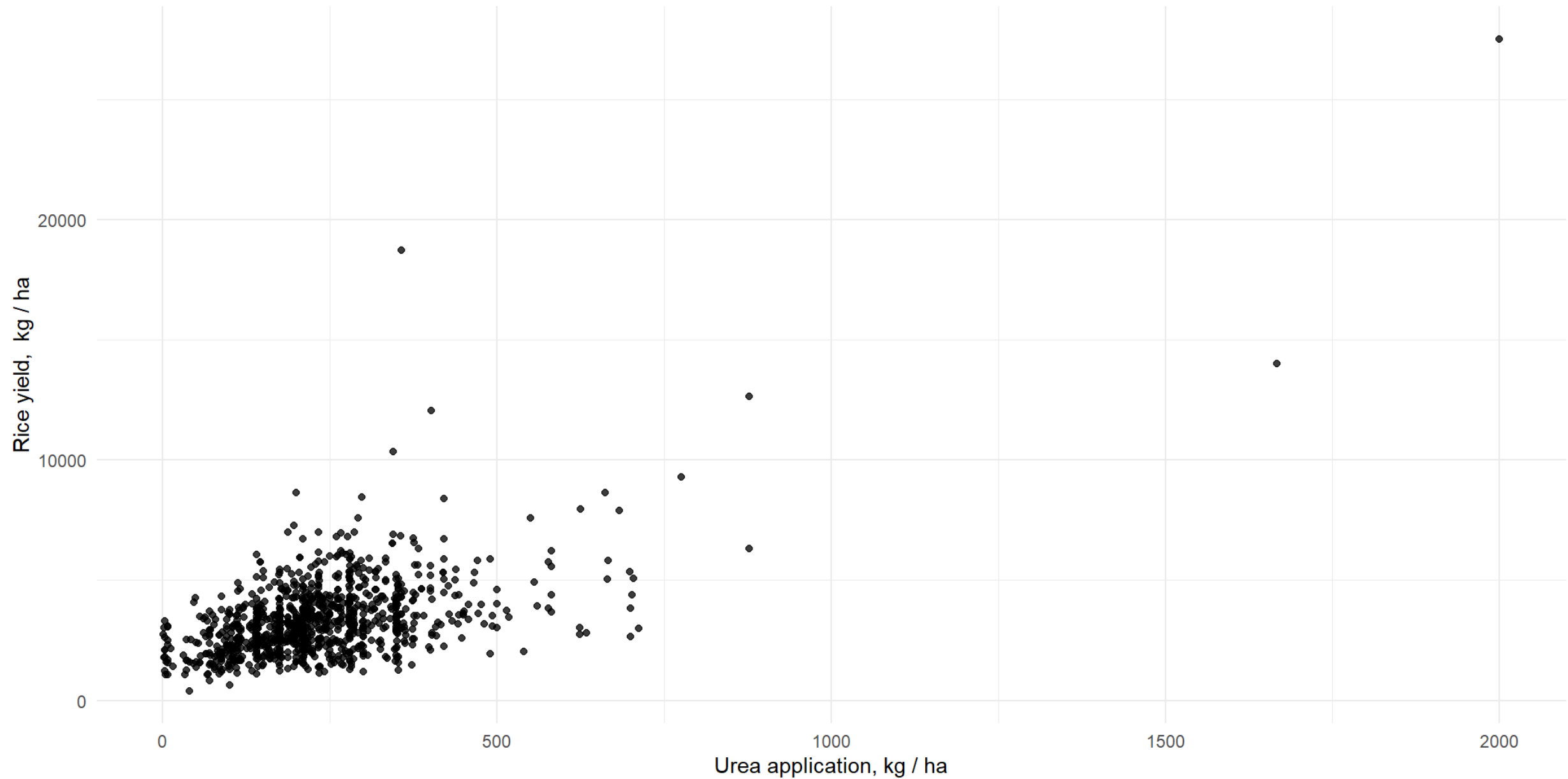
Descriptive statistics for the full sample

Variable	Mean	SD	Median	Min	Max
yield	3381.8	1676.3	3160.0	400.0	27500
urea	236.9	140.1	214.3	0.9	2000

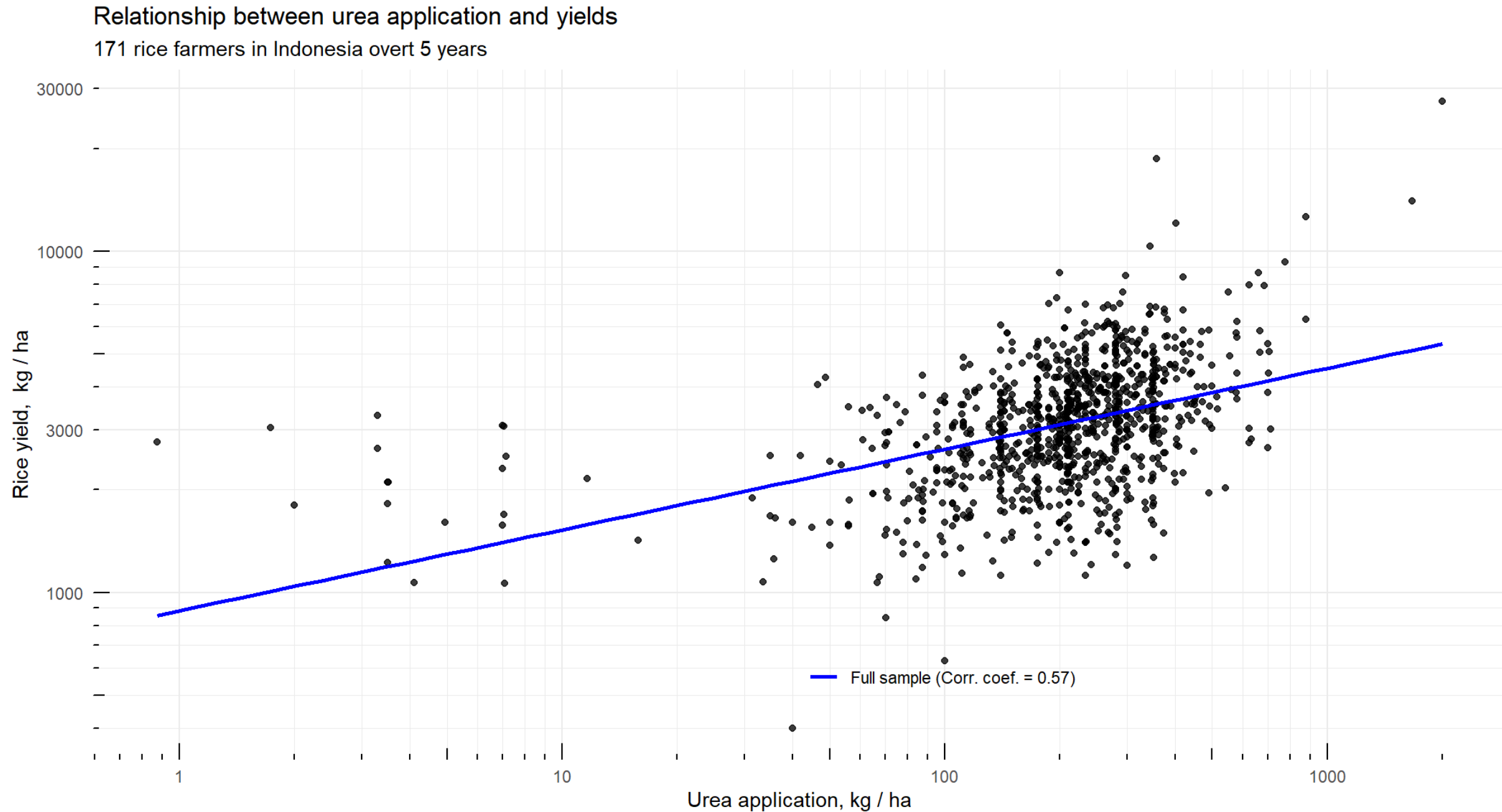
Scatter plot

Relationship between urea application and yields

171 rice farmers in Indonesia over 5 years



Linearity and linear transformation



Correlation (exam questions)

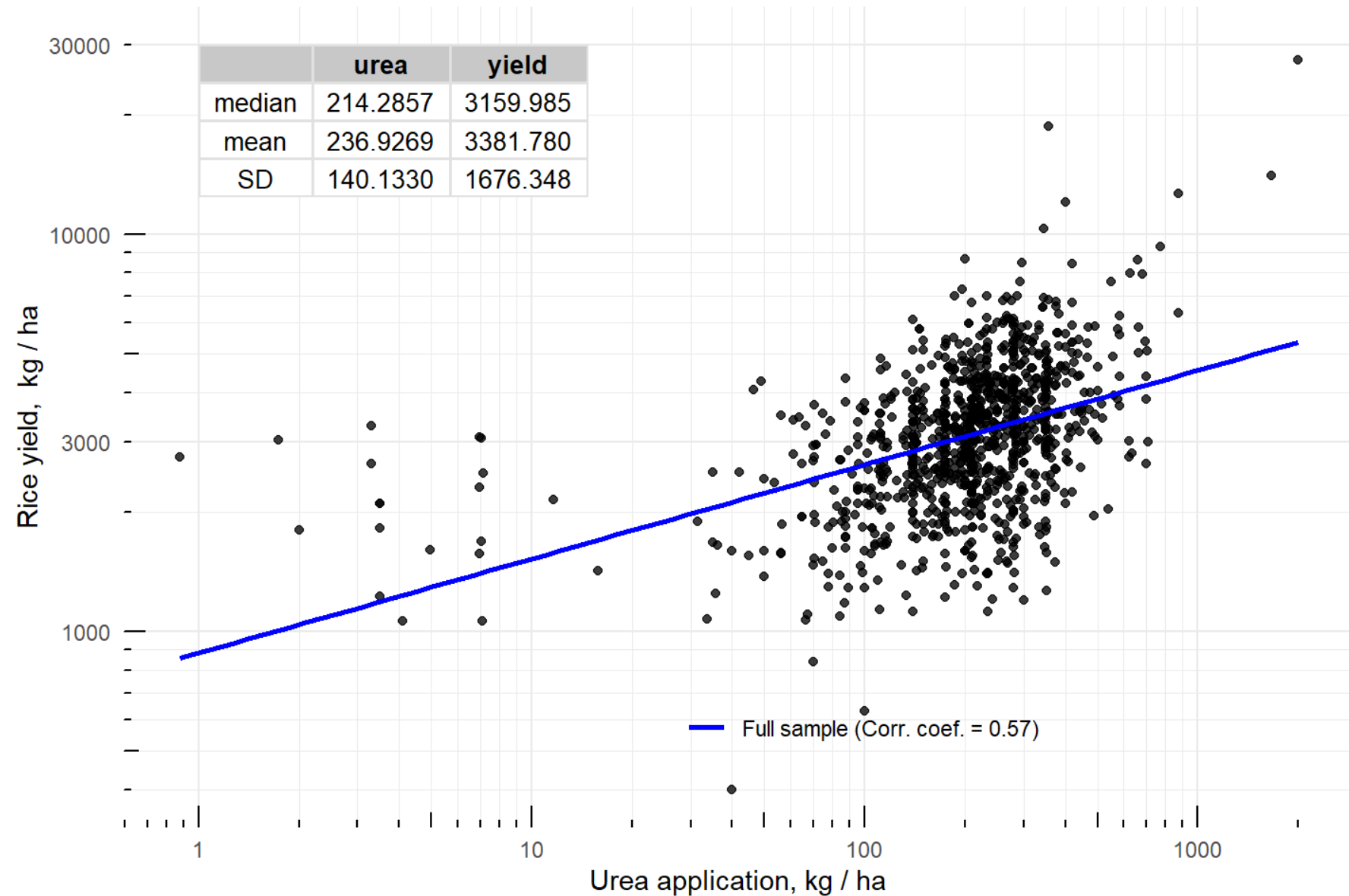
What can we conclude about the relationship?

- Linear / non-linear; strong / weak.
- What is the central tendency and variance?
- What is the effect of outliers on the central tendency and variance?

Does the correlation suggests that urea application causes rice yields increases?

- Explain why

Relationship between urea application and yields
171 rice farmers in Indonesia overt 5 years



Causality conclusion

Does the correlation suggests that urea application causes rice yields increases?

- (Theory) Common knowledge says: fertilizers should increase yields. But:
 - Increase in N without other fertilizers (P and K) may not increase yield because of the
 - **Liebig's law of the minimum.**
 - Water availability, climate (El Niño and La Niña), technology, and other production factors may affect the yields.
 - Soil, seeds, Farmer's skills, motivation, ...
- As data contains different farmers over different years, these factors are likely to be different!

Conclusion:

In the current setting, we **CANNOT** conclude that an increase in application of urea **causes** an increase in rice yields because:

- **other factors are not constant** between different farms/observations/periods. See those factors above.
- In other words, **there is no Ceteris Paribus**

Cause, Effect, Potential Outcome, and the Ceteris Paribus

The econometrician's goal is to :

- infer that one variable (such as fertilizers)
- has a causal effect on another variable (such as yields).

Association between two or more variables might be suggestive, but not causal ([Wooldridge 2020](#)).

Cause and Effect in Rice Yields

1. The Cause or the **Treatment**

- in 2014, farm A uses 100 kg of urea instead of 0: $\text{Urea}_A = 100$

2. Outcome or **Factual**

- rice yields in 2015
- $\text{Yield}_A(\text{Rice}|\text{Urea} = 100)$

3. **Effect**

- How to measure it?

$$\text{Effect} = \text{Outcome} - \text{Potential Outcome}$$

2.a Potential Outcome, or **Counterfactual**

- IF in 2014, farm A did not apply urea;
- potential rice yields in 2015
- $\text{Yield}_A(\text{Rice}|\text{Urea} = 0)$

$$\text{Effect} = \text{Factual} - \text{Counterfactual}$$

$$\text{Causal effect} = \text{Yield}_A(\text{Rice}|\text{Urea} = 100) - \text{Yield}_A(\text{Rice}|\text{Urea} = 0)$$



Once the farm A applied 100 kg of Urea, can we possibly observe the Potential Outcome?

Normally, we observe only factual outcomes:

Farm A

- $\text{Urea}_A = 100$
- $\text{Yield}_A(\text{Rice} | \text{Urea} = 100)$

Farm B

- $\text{Urea}_B = 0$
- $\text{Yield}_B(\text{Rice} | \text{Urea} = 0)$

Normally, we measure **a difference between two factual outcomes:**

$$\text{Yield difference} = \text{Yield}_A(\text{Rice} | \text{Urea} = 100) - \text{Yield}_B(\text{Rice} | \text{Urea} = 0)$$

This is **not the same as the causal effect**, because the other things are not equal between farms A and B.

Cause = Treatment

- Variable / input that
 - we change
 - which effect we want to measure

Outcome

- Measured characteristics that (theoretically) result from a treatment.

Causal Effect

- Difference between **factual** and **counterfactual**
- Difference between factual and potential outcomes.

Potential Outcome / Counterfactual

is an **outcome that would have occurred** if NO treatment had been applied.

- **we cannot observe a counterfactual** because
 - once a treatment is applied, subject cannot be observed without the treatment.
- In counterfactual, all other factors must be equal, except for the treatment:
 - **Ceteris paribus.**

Cause, Effect and the Ceteris Paribus

To measure the causal effect of a treatment, we need to ensure other things equal (ceteris paribus)

The “other things” are:

- **measurable factors** (age, height, costs, ...);
- **measurable but not included in our data;**
- **unmeasurable factors** (ability, effort, connections, ...)

How to establish the ceteris paribus?

There are 5 key econometric methods:

1. **RCT - Randomized Control Trials**
2. **Regression**
3. IV - Instrumental Variable
4. DID - Difference in Difference (Panel Regression)
5. RDD - Regression Discontinuity Design

Ceteris Paribus: Public vs. Private University

Watch and discuss now:

< <https://www.youtube.com/watch?v=iPBV3BlV7jk> >

OR the same [Ceteris Paribus: Public vs. Private University](#)

Randomized Control Trials - RCT

RCT can ensure the Ceteris Paribus!

- When the treatment is assigned randomly in a controlled environment, all other things remain fixed.

Watch now:

- [Randomized Trials: The Ideal Weapon](#)
- or <https://youtu.be/eGRd8jBdNYg>

Randomized Control Trials

Treatment group

- a random sample
- from the population,
- where a **treatment** is assigned.

Control group

- random sample
- from the population,
- where **NO** treatment is assigned.

RCT ensures the Ceteris Paribus The through Law of Large Numbers (LLN):

- The treatment and the control groups are **on average the same**,
- when they are **randomly sampled** from **the same population** with the **large sample size**.
- The only difference between treatment and control is **the fact of treatment**.

Causal effect = Average Causal Effect = Difference in averages between two groups

Ceteris paribus I: RCT (2)

Experiments are often **not available, costly, sometimes unethical**, and often illegal.

Think, about how to make experiments to estimate the effects of:

- years of schooling on earning?
- additional child on mother's employment?
- serving in the army on the job perspective?
- electronics (phones/laptops) in class on students performance?
- health insurance on health?

Ceteris paribus II: Regression

When experiments are not possible, **econometric methods** come to our rescue.

Regression

- **regressions** control the effects of other factors in the **causal effect of interest**.
- **regression** contains **multiple variables**, which **represent measurable and approximate not measurable** factors.
- Regression analysis has to be theoretically sound by ensuring that **key assumptions are validated**.

Ceteris paribus III: Innovation

Use clones!

Add treatment!



Econometrics

Is the science of measuring the effect of a treatment/cause. To learn more:

1. The world of potential outcomes: Ch 1 in ([Joshua D. Angrist and Pischke 2014](#)).
2. It is all about the research design: ([Joshua D. Angrist and Pischke 2010](#)).
3. The RCT is not the ultimate solution in development economics: ([Deaton 2009](#)).
4. More on the RCT in economics: ([Heckman and Smith 1995](#)), ([Burtless 1995](#)).

Example of an RCT (base on Kuchebenbecker et al. (2017))

Research question: Can we reduce children's stunting by training parents about nutrition?

Warning

What RCT experiment can help us answer this question? discuss:

- An outcome;
- A treatment;
- The population;
- A sample;
- How to measure the impact of the treatment?

Example of an experiment

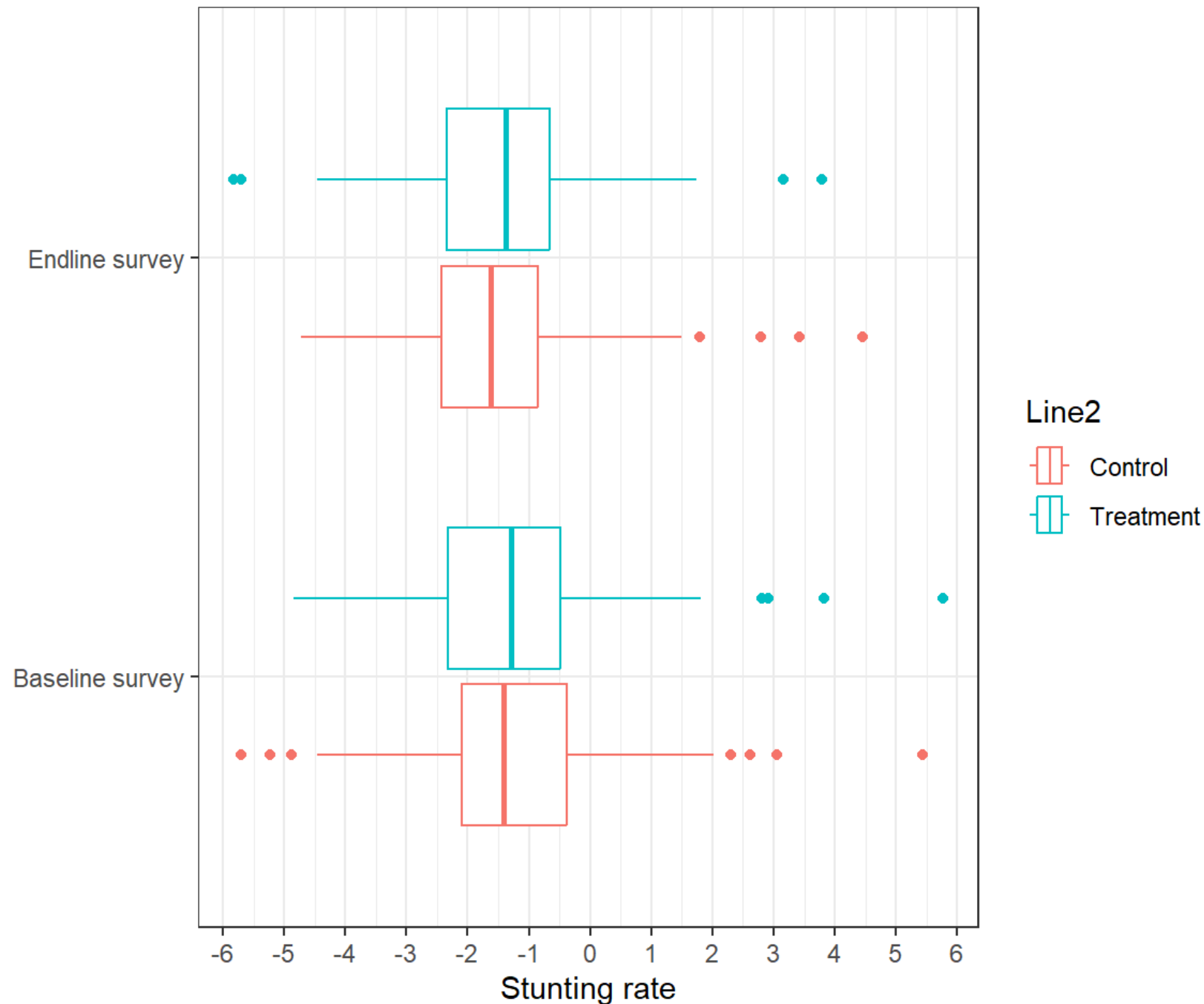
1. An outcome:
 - Stunting score of a child.
2. A treatment:
 - Parents' special training in nutrition.
3. The population
4. A sample:
 - Randomly select eligible subjects.
5. How to measure the impact of the treatment?
 - Randomly assign two groups:
 - treatment (communities with trainings), and
 - control (community without it)
 - Collect the pre-intervention data (baseline)
 - Apply the treatment
 - collect the post-intervention data
 - Measure the effect

Question 1.

Does the **research design** imply that (if found) education has a causal impact on stunting?

- Answer:
 - **Yes!**
 - All other things **may be equal** because the research design is based on the RCT.
 - **Training** is the only change between the treatment and control.

Stunting before-after / treatment-control (1)



Is the average stunting lower in the treatment group compared to the control group?

Answer:

- Hard to say!

Stunting before-after / treatment-control (2)

Line	Stunting score	
	Control	Treatment
Baseline survey	-1.263 [1.490]	-1.245 [1.665]
Endline survey	-1.571 [1.330]	-1.430 [1.455]

The table reports the mean and [SD] of stunting by sample and period.

Is the average stunting lower in the treatment group compared to the control group?

Answer:

- The absolute number, yes. BUT!
- The treatment and control groups are two different samples from the population.
- Means naturally vary in random samples.

Recommendations

Line	Stunting score	
	Control	Treatment
Baseline survey	-1.263 [1.490]	-1.245 [1.665]
Endline survey	-1.571 [1.330]	-1.430 [1.455]

The table reports the mean and [SD] of stunting by sample and period.

Can we recommend education as a way to reduce stunting?

Answer:

- Not yet!
- We must apply statistical inference to draw conclusions about the population based on the estimates.
- If inference confirms the effect, then yes!

Conclusion

To conclude that two means estimated from different samples are also different in the population, we need to use statistical inference!

- **Without statistical inference, we cannot know the likelihood (odds) that the difference in means is due to random variation in the samples.**

Homework

Potential exam questions

- What are the differences between counterfactual and potential outcome?
- What is the counterfactual? How is it different from the factual?
- What can we observe in the data, factual or counterfactual?
- What are the factual and counterfactual in the Urea fertilizers and Rice yields example?
 - How to generate a counterfactual?
- What makes randomized control trials such a powerful tool for revealing the causal effects?
 - What statistical principles ensure the ceteris paribus under the RCT?
- Give examples of measurable and not measurable factors that need to remain fixed in order to reveal the causal effect of fertilizer application on crop yields.

Readings

- Chapter 1 Randomized Trials. In Mastering 'metrics : the path from cause to effect / Joshua D. Angrist and Jörn-Steffen Pischke. Princeton University Press, 2015. <https://justfind.hds.hebis.de/Record/HEB35797798X>

Optional:

- Chapter 1 in “The Book of Why: The New Science of Cause and Effect” by Judea Pearl and Dana Mackenzie
<https://justfind.hds.hebis.de/Record/HEB446798185>

The essence of econometrics (Video)

Econometrics: The Path from Cause to Effect

Key concepts covered in the lecture

Scatter plot:

- How to construct it;
- Linear/non-linear relation between variables
- Variability.

Trend line:

- Linear trend;

Correlation:

- Linear relationship only.
- Interpretation Positive/Negative coefficient
- Non-linearity, outliers and correlation.
- Correlation is not causality.

The Ceteris Paribus:

- Ceteris Paribus;
- Counterfactual and potential outcome;
- Cause and treatment;
- Outcome and Effect;

References

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- Kuchenbecker, Judith, Anika Reinbott, Beatrice Mtimuni, Michael B. Krawinkel, and Irmgard Jordan. 2017. "Nutrition Education Improves Dietary Diversity of Children 6-23 Months at Community-Level: Results from a Cluster Randomized Controlled Trial in Malawi." Edited by Jacobus P. van Wouwe. *PLOS ONE* 12 (4): e0175216. <https://doi.org/10.1371/journal.pone.0175216>.
- Wooldridge, M. Jeffrey. 2020. *Introductory Econometrics: A Modern Approach*. South-Western. <https://www.cengage.uk/shop/isbn/9781337558860>.